The result is one geometric statement. In complex six-dimensional space $\mathbb{C}^6$ there are three mutually unbiased orthonormal bases, and no four. In symbols, $N(6) = 3$.

The proof is long for an honest reason. Some of modern geometry ends in a finite check, and a finite check counts as a proof only when every possible bug, rounding error, and missed case is excluded. So most of the work is *certification*: every test the computer runs carries a strict safety margin, so that ordinary floating-point arithmetic cannot turn a true solution into a spurious "not found."

The strategy fits in six steps.

1. **Remove the first basis.** A unitary change of coordinates sends any orthonormal basis to the standard one. The second basis is then the column set of $H/\sqrt{6}$, where every entry of H has modulus 1 and $HH^* = 6I$. Such H is a *complex Hadamard matrix*, and the question becomes: which H admit two further unbiased bases?

2. **Pin down the phase freedom.** Multiplying rows and columns of H by phases changes nothing. Phases are chosen so that the first row and column of H are 1, after which H is fixed by nine real phase parameters, the rest of the first row and column, together with a 4×4 block of unknown phases.

3. **Cover the seed.** Those nine parameters trace a subset of the nine-torus, and finitely many balls of radius 0.0320001, measured in cycles, cover it. This is the seed cover.

4. **Complete and refine.** For a seed ball, the 4×4 block must satisfy Hadamard equations that compress to a single $V \in U(2)$. Finitely many boxes cover $U(2)$, and each box refines into small phase balls holding the actual solution set. This is the hardest geometry in the argument.

5. **Exclude two further bases.** Take one phase ball for H. Every vector unbiased to both the standard basis and $H/\sqrt{6}$ has the form $e(\beta)/\sqrt{6}$, and a phase-bin search covers them all by small vector balls. Two further bases would need twelve of those balls: six pairwise orthogonal in one group, six in the other, with all 36 cross pairs unbiased. Those conditions are edges of two graphs, and a finite search rules out the required pair of six-cliques.

6. **Account for everything.** The execution of these four stages on a 2026 machine, with a fixed build and fixed numeric margins, is recorded together with the certified margins. The finite chain reaches every possible H up to the allowed symmetries, and each local test either excludes soundly or fails openly. Nothing is guessed.

Formality. All sets of phase coordinates are closed, as are all balls, ellipsoids, and boxes. Overlaps, and repeated overlap with phase representatives, are harmless: duplication weakens an exclusion but never omits a solution. "Small ball" always means the phase matrices within a stated Frobenius norm. A numerical center is a proposal and need not be an exact Hadamard matrix. Every conclusion comes from a necessary-condition inequality verified on a whole ball, not from the residual at a proposed center.

Introduction.

Two orthonormal bases $\mathcal{B}$ and $\mathcal{C}$ of $\mathbb{C}^d$ are *mutually unbiased* (MUB) if

$$|\langle b, c\rangle|^2 = \frac{1}{d} \qquad \text{for every } b \in \mathcal{B},\ c \in \mathcal{C}.$$

Here $\langle u, v\rangle = \sum_j \overline{u_j} v_j$. For $d \geq 2$ let $N(d)$ be the maximum size of a pairwise mutually unbiased family of orthonormal bases in $\mathbb{C}^d$. The main result is:

Theorem 1 (Computer-assisted)**.** *Under the binary64 arithmetic and compiler conditions stated in the certified arithmetic, the execution documented in certifies that there are no four pairwise mutually unbiased orthonormal bases in* $\mathbb{C}^6$*. Consequently* $N(6) = 3$*.*

The upper bound is the hard direction. The lower bound (three bases exist) is an explicit computation, .

MUBs, Hadamard matrices, and the shape of the problem.

Fixing one basis by a common unitary transformation costs nothing; every family can be moved so that the first member is the standard basis. Then the second member is the columns of $H/\sqrt{6}$ for a complex matrix satisfying

$$|H_{ij}| = 1, \qquad HH^* = 6I.$$

Such an H is called a *complex Hadamard matrix*. The issue is to decide whether a given pair (standard basis, $H/\sqrt{6}$) can be extended by up to two more MUB bases, and to do that uniformly over the whole (unclassified) space of complex Hadamard matrices. Because H is square, $HH^* = 6I$ implies $H^*H = 6I$ as well.

The answer we prove, Theorem 1, is that no such extension exists, for any H. The proof covers the full space of H, not just a known family: the second basis is completely unconstrained, and the argument handles complex Hadamard matrices outside all previously studied families.

Historical context.

Mutually unbiased bases describe projective measurements in which a state drawn from one basis gives a uniformly random outcome in any other. Schwinger placed this complementarity inside his calculus of unitary operator bases in 1960 (Proc. Nat. Acad. Sci. USA 46, 570–579). Ivanović raised the state-determination problem and gave the prime-dimensional construction in 1981 (J. Phys. A 14, 3241–3245). Wootters and Fields proved that a complete set of such bases is statistically optimal for state determination and built one whenever the dimension is a prime power (Ann. Physics 191, 363–381, 1989). Every dimension obeys $N(d) \leq d + 1$, and the short operator-space proof is due to Bandyopadhyay, Boykin, Roychowdhury and Vatan (Algorithmica 34, 512–528, 2002).

https://www.pnas.org/doi/10.1073/pnas.46.4.570

https://doi.org/10.1088/0305-4470/14/12/019

https://www.sciencedirect.com/science/article/pii/0003491689903229

https://arxiv.org/pdf/quant-ph/0103162

Dimension six is the first case that is neither a prime power nor a small even dimension settled by other constructions. A tensor-product construction already yields three bases in $\mathbb{C}^6$, tensoring a qubit pair with a qutrit triple; that is Section 4 of the Klappenecker and Rötteler study of MUB constructions. Zauner conjectured in his 1991 Vienna Diplomarbeit that three is the maximum, the same statement reappears in his doctoral thesis, and it is Conjecture 1 of the Klappenecker–Rötteler paper. A survey of the composite-dimension setting is due to McNulty and Weigert.

https://arxiv.org/pdf/quant-ph/0309120

https://www.gerhardzauner.at/documents/gzdiplthd.pdf

https://arxiv.org/pdf/2410.23997

Four attacks ran alongside. Grassl proved that the standard basis together with any member of the two-parameter Fourier family cannot extend to four bases, Theorem 2 of his dimension-six note. Bengtsson, Bruzda, Ericsson and Życzkowski mapped the order-six Hadamard families and the unbiased vectors they carry. Butterley and Hall, and independently Brierley and Weigert, produced the numerical evidence that no four bases exist. Brierley and Weigert also enumerated the unbiased vectors of individual Hadamards through algebraic equations: Section 3.4 of that work gives rigorous approximation bounds for exact input matrices, and Section 4.3 explains why rounding coefficients

in sampled non-affine families blocks a complete cover by that route. Raynal, Lu and Englert sharpened the numerical side by maximising the average distance between four bases and finding the maximum strictly below unity, with a two-parameter family of three bases attaining it.

https://arxiv.org/pdf/quant-ph/0406175
https://arxiv.org/pdf/math.OA/0609076
https://arxiv.org/pdf/0901.4051
https://arxiv.org/pdf/1103.1025

The closest antecedent is the certified discretization of Jaming, Matolcsi, Móra, Szöllősi and Weiner. They proved that the standard basis together with any member of the two-parameter Fourier family cannot extend to four bases, Theorem 1.4 of their 2009 paper, and their Section 3 introduces closed phase intervals, rigorous error bounds, subdivision, and exhaustive compatibility tests. Jaming, Matolcsi and Móra then set out the unrestricted discretization programme in Sections 2–4 of their survey. Those four devices carry the finite search throughout: what is added is exhaustive coverage of arbitrary second bases, with every error bounded.

https://arxiv.org/pdf/0902.0882
https://arxiv.org/pdf/1201.0640

Two recent lines of work are complementary rather than overlapping. Matolcsi, Matszangosz, Varga and Weiner study triplets of MUBs under conjectural sufficient conditions, Section 4 of their preprint. Cárdenes Wuttig and Tindall classify order-six Hadamard matrices completely and leave MUB extendibility open. Only the combination of a finite cover of all second bases with the local exclusion, executed under a certified arithmetic contract, yields an unconditional upper bound.

https://arxiv.org/pdf/2503.14752
https://arxiv.org/pdf/2608.18053

The strategy at a glance.

The proof has five layers, each of which strictly refines the previous one and preserves every exact candidate solution.

1. **Seed cover.** Row and column rephasings and transpositions place any H into a standard form with 9 free phases. A symmetry reduction plus necessary interval tests cover the possible values of these nine phases by finitely many seed balls (radius 0.0320001 in cycle units).

2. **Completion.** In a block decomposition of H with row and column sizes $2 + 4$, the seed pins down the upper-left, upper-right and lower-left blocks. The lower-right 4×4 block is the only remaining unknown. At any seed center that passes the frame checks, the upper-right block and the conjugate transpose of the lower-left block have rank two, and so they admit two-dimensional kernels in $\mathbb{C}^4$. Compressing the unknown lower-right block between these kernels produces a 2×2 matrix whose polar part is a unitary $V \in U(2)$. A closed cover of $U(2)$ therefore accounts for every feasible completion, and uniform Taylor bounds within each cell either exclude the cell or place all its exact completions in a certified phase ball (and).

3. **Refinement.** Each phase ball carries numerical tangent directions and a rigorous residual-identity estimate; covering the tangent coordinates while keeping the residual bound yields smaller balls that still cover every exact solution. No manifold parametrization is used, and the estimates remain valid even when the proposed center is not an exact Hadamard matrix.

4. **Exclusion.** For a final phase ball, a small phase-bin search covers every vector unbiased to both the standard basis and $H/\sqrt{6}$ by closed vector balls. These vector balls become vertices of two graphs, one for orthogonality and one for unbiasedness. Necessary numerical

tests remove only impossible edges. Two additional MUB bases would require two disjoint six-cliques with all 36 cross edges present; that pattern is ruled out by finite search.

5. **Execution certificate.** A complete run of the four programs, on a specific build and platform, accounts for every seed, every refined ball, and every initially unresolved record. Nothing is left open: a record that is not excluded must be covered upstream, and every fallback record must produce a negative exclusion.

This restates the same six-bullet story from the preface, now with the section labels attached. If you only remember one thing, remember this: the argument is a chain of strictly finer covers; each exclusion uses only a *necessary* condition with a *certified* margin; and no step in the chain ever relies on a floating-point center being exact.

The elementary lower bound.

The lower bound uses only the tensor-product construction. We give it explicitly so the reader can see that attainment of three bases is separate from the certified exclusion of four.

Lemma 2 (Three mutually unbiased bases in dimension six)**.** *There exist three pairwise mutually unbiased orthonormal bases in* $\mathbb{C}^6$*.*

Proof. Identify $\mathbb{C}^6$ with $\mathbb{C}^2 \otimes \mathbb{C}^3$ in the standard product coordinates. Define

$$A_0 = I_2, \qquad A_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad A_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}.$$

These are unitary, and for $r \neq s$, every entry of $A_r^* A_s$ has squared modulus $1/2$; the only cross product not immediate from the entries is

$$A_1^* A_2 = \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}.$$

Let $\omega = e^{2\pi i/3}$ and set

$$F = \frac{1}{\sqrt{3}} \big(\omega^{jk}\big)_{j,k=0}^{2}, \qquad D = \operatorname{diag}(1, \omega, \omega), \qquad B_0 = I_3,\ B_1 = F,\ B_2 = DF.$$

The relation $1 + \omega + \omega^2 = 0$ makes F unitary; D is unitary as well. Both F and DF have entries of squared modulus $1/3$. For their cross product,

$$(F^* DF)_{jk} = \frac{1}{3} \sum_{x=0}^{2} \omega^{x^2 + (k-j)x}.$$

If $t = k - j$ is read modulo 3, the sum is $1 + 2\omega$ for $t = 0$ and $2 + \omega^2$ for $t = 1, 2$. Each has squared modulus 3. Consequently every entry of $B_r^* B_s$ has squared modulus $1/3$ when $r \neq s$.

For $r = 0, 1, 2$, let the columns of $U_r = A_r \otimes B_r$ form the rth basis. Each U_r is unitary. For $r \neq s$ we have $U_r^* U_s = (A_r^* A_s) \otimes (B_r^* B_s)$, so every entry has squared modulus $(1/2)(1/3) = 1/6$. Thus the three bases are mutually unbiased, and $U_0 = I_6$. □

A Lie-algebra consequence.

An orthonormal basis $\mathcal{B}$ of $\mathbb{C}^6$ determines its Cartan subalgebra $\mathfrak{h}_{\mathcal{B}}$ of traceless matrices that are diagonal in that basis. It is spanned by the matrices $bb^* - I/6$ for $b \in \mathcal{B}$. The Killing form on

$\mathfrak{sl}_6(\mathbb{C})$ is $K(X,Y) = 12\,\mathrm{tr}(XY)$; two subalgebras are *Killing-orthogonal* when this form vanishes on every pair of elements. For unit vectors b, c,

$$\mathrm{tr}((bb^* - I/6)(cc^* - I/6)) = |\langle b, c\rangle|^2 - \frac{1}{6}.$$

So two bases are mutually unbiased exactly when their associated Cartan subalgebras are Killing-orthogonal. These algebras are closed under the standard adjoint, and the converse correspondence also uses this adjoint condition.

Corollary 3 (Adjoint-closed Cartan subalgebras)**.** *Under the arithmetic and compiler conditions of the certified arithmetic, using the complete execution recorded above and certified in Theorem 1, the maximum number of pairwise Killing-orthogonal Cartan subalgebras of $\mathfrak{sl}_6(\mathbb{C})$ that are all closed under the standard adjoint $X \mapsto X^* = \overline{X}^T$ is three.*

Proof. By Theorem 5.2 of Boykin, Sitharam, Tiep and Wocjan, such collections of Cartan subalgebras correspond one-to-one to collections of mutually unbiased orthonormal bases of the same cardinality. Theorem 1 gives the upper bound under its stated contract, and Lemma 2 gives attainment. □

https://arxiv.org/pdf/quant-ph/0506089

Equivalences and an exhaustive seed cover.

The first layer of the cover replaces every possible second basis by one of finitely many nine-phase seed balls. The strategy, closed phase discretization plus necessary compatibility tests, follows Section 3 of Jaming, Matolcsi, Móra, Szöllősi and Weiner. The unrestricted programme is set out in Sections 2–3 of the later survey by Jaming, Matolcsi and Móra. What is new is the specific symmetry reduction and shared-coordinate estimate that prove completeness of this exact cover.

https://arxiv.org/pdf/0902.0882

https://arxiv.org/pdf/1201.0640

Throughout, phases are in cycles and $e(t) = \exp(2\pi i t)$ entrywise. Matrix indices run $0, \ldots, 5$, entries are stored in row-major order with (i, j) in position $6i + j$, and the Hermitian inner product is $\langle u, v\rangle = \sum_j \overline{u_j} v_j$. Vectors and phase matrices use Euclidean and Frobenius norms respectively; operator norms are induced Euclidean unless noted. An angle is in radians only when explicitly stated.

Equivalences and phase gauges.

Definition 4. A matrix $H \in \mathbb{C}^{6\times 6}$ is *Hadamard* if every entry has modulus one and $HH^* = 6I$. It is *extendible* if the standard basis and the columns of $H/\sqrt{6}$ belong to a family of four mutually unbiased orthonormal bases.

Unbiasedness to the standard basis forces every entry of the second basis matrix to have modulus $1/\sqrt{6}$; the matrix is then $H/\sqrt{6}$ with H Hadamard. Because H is square, $HH^* = 6I$ implies $H^*H = 6I$. Every such H has a real phase lift P with $H = e(P)$.

Lemma 5 (Equivalences)**.** *Extendibility is invariant under row or column permutations, multiplication of rows or columns by unit scalars, complex conjugation, and transposition.*

Proof. Permuting or rephasing columns only reorders or rephases vectors within the second basis. A row permutation or row rephasing is a common unitary operation on all four bases; it sends the standard basis to a new basis but preserves both orthonormality and the moduli of inner products among the four bases.

For transposition, write $U = H/\sqrt{6}$ and apply the common unitary U^* to a family whose first two basis matrices are I, U. Those two become U^*, I; interchange their roles and conjugate every coordinate. The first two matrices are now I, U^T. Every operation is reversible, so the statement holds in both directions. □

For a real 6×6 matrix z define double centering by

$$(\Pi z)_{ij} = z_{ij} - \frac{1}{6}\sum_b z_{ib} - \frac{1}{6}\sum_a z_{aj} + \frac{1}{36}\sum_{a,b} z_{ab}. \tag{1}$$

Its range is the *horizontal* subspace of matrices whose every row and column sum is zero. We also use unwrapped dephasing

$$D_0(P)_{ij} = P_{ij} - P_{i0} - P_{0j} + P_{00}. \tag{2}$$

Lemma 6 (Horizontal phase representatives)**.** Π *is the orthogonal projection onto the horizontal subspace; in particular* $\|\Pi z\|_F \le \|z\|_F$*. If two phase representatives differ by* z*, the second is equivalent under row and column rephasings to the first plus* Πz*. The statement survives arbitrary entrywise integer changes of phase representatives.*

Proof. Equation (1) has zero row and column sums, and $z - \Pi z$ is a sum of a matrix constant on each row and a matrix constant on each column. Every such sum is orthogonal to the horizontal subspace in the real Frobenius inner product, so Π is orthogonal projection onto that subspace. If $Q = P + z$, then $P + \Pi z = Q - (z - \Pi z)$ differs from Q by row and column phases, allowed by Lemma 5. Integer additions leave $e(P)$ unchanged. □

Similarly $D_0(P)$ gives an equivalent matrix whose first row and column are zero. If a wrapped difference is used, its chosen integers determine a legitimate phase representative; no rounded decision at a half-integer is claimed to be uniquely preferred.

A symmetry arrangement for nine phases.

The *gap* of a finite circular multiset is its largest circular gap, equivalently the length of an open arc that contains no point of the multiset. Repeated points count with multiplicity; a repeated point contributes a zero gap. For every pair of distinct rows of P consider their six phase differences modulo one, and similarly for columns.

Lemma 7 (Seed arrangement)**.** *Every exact Hadamard matrix has an equivalent dephased representative for which a pair of rows attaining the minimum gap among all row and column pairs is the first pair, and its phases can be written*

$$x = (0, x_1, x_2, x_3, x_4, x_5), \quad 0 \le x_1 \le \cdots \le x_5 \le 1, \quad g_x = 1 - x_5, \quad x_1 \le x_5 - x_4. \tag{3}$$

After any one of these five noninitial columns is selected and moved to position two, the last four rows can be permuted so that the second column has phases

$$y = (0, \xi, y_2, y_3, y_4, y_5), \qquad 0 \le y_2 \le y_3 \le y_4 \le y_5 \le 1, \tag{4}$$

where ξ *is the selected entry of* x*. These lists satisfy*

$$\sum_{j=0}^{5} e(x_j) = 0, \qquad \sum_{j=0}^{5} e(y_j) = 0, \qquad g_y \ge g_x. \tag{5}$$

Proof. There are finitely many row and column pairs, so a pair of minimum gap exists. Transpose if it is a column pair, then permute rows so it is the first pair. Choose one of its largest gaps, start the circular ordering immediately after that gap, and use that column as the first column. Dephase the first row and column. Ordering the remaining columns along the complementary arc gives $g_x = 1 - x_5$ and the sorted list in (3).

If the final orientation inequality fails, interchange the two selected rows, use the old x_5 column as the starting column, and reverse the ordering. The new list is

$$x'_0 = 0, \qquad x'_i = x_5 - x_{5-i} \quad (1 \le i \le 5).$$

It has the same distinguished largest gap, and $x'_1 = x_5 - x_4$ whereas $x'_5 - x'_4 = x_1$. So this orientation satisfies the required inequality whenever the other fails. The construction uses a chosen largest gap and is unchanged by ties or repeated points.

Select a noninitial column and place it at position two. Permuting the last four rows sorts its last four entries without changing either of the first two rows. Orthogonality of those rows gives the first vanishing sum in (5); orthogonality of the first two columns gives the second. Row and column rephasings rotate the relevant difference multisets; permutations merely reorder points or relabel pairs. Transposition interchanges the two collections of pairs, so the minimum gap stays g_x and the selected column pair has $g_y \ge g_x$. □

Set $N = 112$ and $w = 1/N$. Use closed bins

$$I_b = [b/N, (b+1)/N], \qquad u_b = (b + \tfrac{1}{2})/N, \qquad 0 \le b < N. \tag{6}$$

They cover $[0, 1]$ including both endpoints. Sorted exact phases admit nondecreasing bin indices, so write the five indices for $x_1, \dots, x_5$ as $b^X = (a, b, c, d, l)$ with $0 \le a \le b \le c \le d \le l < N$. The orientation condition implies the necessary integral restriction

$$\frac{a}{N} \le x_1 \le x_5 - x_4 \le \frac{l + 1 - d}{N}, \qquad a \le l - d + 1. \tag{7}$$

For this index tuple choose $s \in \{0, \dots, 4\}$ as the first minimizer of $|2b^X_s + 1 - N|$, the rule used in `seeds::mk`. The selected original column is $s + 1$, the selected exact phase is $\xi = x_{s+1}$, and the new column order is: original column zero, column $s + 1$, and the other four columns in their former order. This specifies only a column; it imposes no extra condition on ξ.

After arranging y as in (4) its five varying entries have bins $b^Y = (b^X_s, b^Y_1, b^Y_2, b^Y_3, b^Y_4)$ with $0 \le b^Y_1 \le b^Y_2 \le b^Y_3 \le b^Y_4 < N$. The shared entry uses the same bin as in x; it is not required to be ordered relative to the four tail bins.

Lemma 8 (Stability of circular gaps)**.** *If each point of one circular multiset is matched to a point of another at circular distance $\le \delta$, their gaps differ by $\le 2\delta$.*

Proof. Take an empty open arc of length L in the first multiset. If $L > 2\delta$, trim δ from each end; a point of the second multiset inside the remaining open arc would have its matched first point in the original empty arc, contradiction. Hence the second gap is at least $L - 2\delta$. If $L \le 2\delta$ the same bound follows from nonnegativity. Apply to a largest gap and interchange the two multisets. No preservation of sorted order is used. □

Lemma 9 (Necessary midpoint gap filters)**.** *For every bin pair containing lists as in Lemma 7, the midpoint gaps (with the fixed zero) satisfy*

$$g_{x,\mathrm{mid}} \le 1 - \frac{l-1}{N}, \qquad g_{y,\mathrm{mid}} \ge 1 - \frac{l+2}{N}. \tag{8}$$

Proof. Each varying phase is within $w/2$ of its midpoint and the initial zero is unchanged. Lemma 8 bounds each gap change by w. Since $x_5 \in [l/N, (l+1)/N]$ we have $1-(l+1)/N \le g_x \le 1-l/N$. Combine with $g_y \ge g_x$. □

For sorted midpoint bin indices $q_0, \dots, q_4$ the gap used by `mk` is

$$\max\Big\{\frac{q_0+\frac{1}{2}}{N},\ \frac{q_1-q_0}{N}, \dots, \frac{q_4-q_3}{N},\ 1-\frac{q_4+\frac{1}{2}}{N}\Big\}.$$

For a Y tuple it sorts a copy solely for this computation, preserving the stored shared-first ordering.

Vanishing-sum filters and the shared displacement.

For either of the two lists x or y, let $u = (u_0, \dots, u_4)$ be its five varying midpoints. Define real two-vector c and real 2×5 matrix J by

$$c = \frac{1 + \sum_{j=0}^{4} e(u_j)}{2\pi i}, \qquad J_j = e(u_j), \tag{9}$$

realified in both definitions.

Lemma 10 (Necessary sum test)**.** *If* $1 + \sum_{j=0}^{4} e(u_j + \Delta_j) = 0$ *and* $|\Delta_j| \le w/2$ *then*

$$\|c + J\Delta\| \le \frac{5\pi w^2}{4}. \tag{10}$$

In particular for every $v \in \mathbb{R}^2$*,*

$$|v \cdot c| \le \frac{5\pi w^2}{4}\|v\| + \frac{w}{2}\sum_{j=0}^{4} |v \cdot J_j|. \tag{11}$$

Proof. The second derivative of $e(t)$ has modulus $4\pi^2$, so Taylor's formula with integral remainder gives

$$\Big|\frac{e(u_j+\Delta_j) - e(u_j)}{2\pi i} - e(u_j)\Delta_j\Big| \le \pi\Delta_j^2.$$

Sum the five remainders and use the vanishing-sum hypothesis to obtain (10). Dot with v, apply the triangle inequality and $|\Delta_j| \le w/2$ to get (11). □

The function `sumok` tests (11) in the residual direction and in directions perpendicular to the five unit-exponential columns. Since the inequality holds for every real direction, their numerical construction need not reproduce any prescribed exact direction. One-sided margins for evaluating the residuals and supports are established in the certified arithmetic.

In grid units $h = N\Delta$, write

$$E = \frac{5\pi}{4N}, \qquad \|Jh + Nc\| \le E, \qquad |h_j| \le \tfrac{1}{2}. \tag{12}$$

Lemma 11 (Shared-coordinate intervals)**.** *Let t be the position of a shared entry in either list. Every feasible displacement satisfies, for every $v \in \mathbb{R}^2$,*

$$|Nv \cdot c + (v \cdot J_t)h_t| \le E\|v\| + \frac{1}{2}\sum_{j \ne t} |v \cdot J_j|. \tag{13}$$

If $a = v \cdot J_t \ne 0$ and the right side is r, this is the interval constraint

$$-\frac{Nv \cdot c}{a} - \frac{r}{|a|} \le h_t \le -\frac{Nv \cdot c}{a} + \frac{r}{|a|}. \tag{14}$$

The intervals obtained for the shared entry from X and Y must intersect.

Proof. Separate the shared term in (12), take its dot product with v, and bound each remaining coordinate by $1/2$. Solving the resulting absolute-value inequality gives (14) for either sign of a. The shared midpoint is identical in both lists, so the same exact grid displacement satisfies both sets of necessary intervals. □

The function `extend` begins with $[-1/2, 1/2]$ and intersects these constraints using the shared column direction and directions perpendicular to the other four columns; here $t = s$ for X and $t = 0$ for Y. It skips divisors of magnitude below 10^{-6}, which only removes a constraint. Numerator and endpoint margins enlarge the computed intervals (the certified arithmetic). The pair is rejected only when the two intervals are strictly disjoint; a common endpoint is retained.

The enumeration in `seeds::build` now has an unambiguous meaning. It visits every weakly increasing five-bin X tuple and applies (7), the first inequality in (8), and `sumok`. Separately, for every possible first bin, it visits all weakly increasing four-bin Y tails and applies `sumok`. The program `covout` matches the shared bin, applies the second gap inequality, and intersects the shared displacement intervals. So every discarded exact solution would violate one of these necessary conditions. Numerical gap comparisons include outward slack (the certified arithmetic).

Certified reuse of seed centers.

The nine seed coordinates, before moving the selected column, are

$$q = (x_1, x_2, x_3, x_4, x_5, y_2, y_3, y_4, y_5) \in \mathbb{R}^9. \tag{15}$$

For a bin pair let $u \in \mathbb{R}^9$ be its rational midpoint and put $h = N(q - u)$. Thus $h \in [-1/2, 1/2]^9$; the shared displacement is h_s. Retaining u as a center would cover the whole bin with radius $3/(2N) < 0.032$. To reduce the number of retained centers, we instead test whether the feasible portion lies in a radius-0.032 ball about a previously retained midpoint. If no such test succeeds we retain u itself.

Let J_X, J_Y, c_X, c_Y be the data in (9). Define the real 4×9 matrix J_* by its action

$$J_* h = \begin{pmatrix} J_X(h_0, h_1, h_2, h_3, h_4)^T \\ J_Y(h_s, h_5, h_6, h_7, h_8)^T \end{pmatrix}, \qquad p_* = \begin{pmatrix} -Nc_X \\ -Nc_Y \end{pmatrix}. \tag{16}$$

For feasible h we have $J_* h = p_* + r_*$, where the two real two-vector blocks r_X, r_Y of r_* satisfy $\|r_X\|, \|r_Y\| \le E$.

Lemma 12 (Support bounds for center reuse)**.** *Let u' be a previously retained midpoint with the same selected position s, and let $k = N(u - u') \in \mathbb{Z}^9$. Every feasible displacement in the current bin pair satisfies*

$$N^2\|q - u'\|^2 = \|k + h\|^2 \le \|k\|^2 + \frac{9}{4} + 2k \cdot h. \tag{17}$$

Two upper bounds for the last term are

$$2k \cdot h \le \sum_{j=0}^{8} |k_j|, \tag{18}$$

$$2k \cdot h \le \sum_{j=0}^{8} |k_j - (J_*^T v)_j| + 2v \cdot p_* + 2E(\|v_X\| + \|v_Y\|), \qquad v \in \mathbb{R}^4, \tag{19}$$

where v_X, v_Y are the two real two-vector blocks of v.

Proof. Expand the square in (17) and use $\|h\|^2 \le 9/4$. The coordinate bounds give (18). For the other bound write

$$2k \cdot h = 2(k - J_*^T v) \cdot h + 2v \cdot p_* + 2v \cdot r_*.$$

The first term is at most $\sum_j |k_j - (J_*^T v)_j|$ by the same coordinate bound. Cauchy–Schwarz on the two residual blocks bounds the last term by $2E(\|v_X\| + \|v_Y\|)$. □

The function `supdual` proposes vectors v and evaluates the right side of (19). Validity of a proposal is independent of how it was generated and of the accuracy of the heuristic linear solve; only finite components of magnitude at most 10^4 are used. Its evaluation replaces $1.25\pi/N$ by the larger $1.251\pi/N$ and adds a final support margin; the certified arithmetic proves the resulting bound is conservative.

Set $R = 0.032$. For a proposed previous center, `testBucket` first tries (18) and then, if applicable, the dual bound. It returns success only when the upper bound in (17) is below the safely lowered comparison threshold for $(NR)^2$. Success therefore certifies the entire feasible portion of the current bin pair, not merely its midpoint. Distance cutoffs, bucket indices, and reach tests only decide which previous centers to try; omitting a possible previous center cannot produce a false success.

For clarity: the shared column is counted once in (15), though its displacement enters both equations in (16). Reuse requires equality of the selected position s. Moving that position to the start of the five X coordinates applies the same coordinate permutation to the current point and to the retained center, and preserves their distance. The resulting order is: the selected X entry, the other four X entries in order, and the four Y tail entries. These are exactly the nine entries of the dephased first two rows and columns used by the next stage. Centers retained earlier for the same X tuple are handled identically, with the first five components of k zero.

The exhaustive seed cover proposition.

Proposition 13 (Exhaustive seed cover)**.** *For a completed execution of the specified* `covout` *traversal under the arithmetic conditions of the certified arithmetic, the written rational midpoint centers, equipped with their recorded selected positions, cover the seed coordinates of every exact complex Hadamard matrix up to the equivalences of Lemma 5, by closed Euclidean balls of radius* 0.032*. After the indicated column ordering and the conversion of centers to binary phase values, radius* 0.0320001 *suffices.*

Proof. Given an exact Hadamard matrix, first apply Lemma 7 to arrange x. Choose a nondecreasing closed bin assignment, determine s from that tuple, and then arrange and assign y with the same shared bin. All these choices exist on bin boundaries and at repeated phases. The tuple appears in the stated enumeration. Lemmas 9, 10, and 11, together with their certified outward numerical margins, ensure that the bin pair is not rejected.

Consider the retained centers as the remaining bin pairs are traversed. If a coverage test succeeds, Lemma 12 and the certified margins put every feasible point of the current pair in a radius-R ball

about an already retained center. If no test succeeds, the program retains the current midpoint itself. Its distance from any point of its full nine-dimensional bin is at most

$$\frac{\sqrt{9}}{2N} = \frac{3}{224} < 0.032.$$

Thus, by induction over the traversal, the retained centers cover every feasible point of every processed pair. Restricting which previous centers are tried can only cause further midpoint insertions; retained centers remain and are included in the output.

The common selected-position permutation preserves all these distances. the certified arithmetic bounds the joint error of the nine binary midpoint phases by less than 10^{-14}, so the triangle inequality permits the enlarged radius 0.0320001. The centers need not themselves satisfy the exact Hadamard equations; their balls cover the necessary seed coordinates of all exact solutions. □

Completion over a fixed seed ball.

This section reduces "complete the seed phase ball to a full dephased Hadamard matrix" to a covering of the 2×2 unitary group $U(2)$. The reference frames are fixed at the seed center; the center itself is not assumed to be an exact Hadamard matrix. All statements are first proved in exact real and complex arithmetic; the certified arithmetic replaces each inequality by an outward-computed version with a certified margin.

Seed blocks and fixed complementary frames.

Fix the stored seed center and take its binary phase coordinates as exact real numbers. Move the shared column into position two, write its relevant unit entries as $a, b_0, b_1, b_2, b_3, c_0, c_1, c_2, c_3$, and put

$$E_0 = \begin{pmatrix} 1 & 1 \\ 1 & a \end{pmatrix}, \qquad B_0 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ b_0 & b_1 & b_2 & b_3 \end{pmatrix}, \qquad (C_0)_{i,:} = (1, c_i) \quad (0 \le i < 4). \tag{20}$$

A feasible completion is a unit-modulus matrix

$$H' = \begin{pmatrix} E' & B' \\ C' & D' \end{pmatrix}, \qquad H'H'^{*} = H'^{*}H' = 6I,$$

with first row and first column all one. Its seed displacements, in radians, are

$$s = (s_a, s_{b_0}, \ldots, s_{b_3}, s_{c_0}, \ldots, s_{c_3}) \in \mathbb{R}^9;$$

and the varied entries are ae^{is_a}, $b_j e^{is_{b_j}}$, $c_i e^{is_{c_i}}$. The shared entry appears exactly once in s. In zero-based row-major phase indexing these positions are

$$(7, 8, 9, 10, 11, 13, 19, 25, 31). \tag{21}$$

Let r_s be the enlarged seed radius in cycles and $R_s = 2\pi r_s$; in the stated computation $r_s = .0320001$. Put $\Delta E = E' - E_0$, similarly $\Delta B, \Delta C$, and abbreviate $e_1 = \|\Delta E\|_F$, $b_1' = \|\Delta B\|_F$, $c_1' = \|\Delta C\|_F$. The chord bound $|e^{ix} - 1| \le |x|$ gives

$$e_1^2 + (b_1')^2 + (c_1')^2 \le \|s\|^2 \le R_s^2. \tag{22}$$

Lemma 14 (Fixed frames)**.** *Let $p \in \mathbb{C}^4$ have unit-modulus entries, $S = \sum_j p_j$, and $M = (\mathbf{1}\ p)$. The squared singular values of M are $4 \pm |S|$. If $|S| < 4$ then*

$$M(M^*M)^{-1} = \frac{1}{16 - |S|^2}(4\mathbf{1} - p\overline{S}\,,\ 4p - \mathbf{1}S). \tag{23}$$

Under the same condition there is an orthonormal 4×2 matrix K with $\operatorname{range} K = \ker M^*$; it can be constructed by completing $\mathbf{1}/2$ and the normalized residual of p with projected coordinate vectors having nonzero residuals.

Proof. $M^*M = \begin{pmatrix} 4 & S \\ \overline{S} & 4 \end{pmatrix}$, whose eigenvalues and inverse give the singular-value formula and (23). The first two Gram–Schmidt vectors span $\operatorname{ran} M$. At each subsequent step the orthogonal complement of the chosen vectors is nonzero; since the coordinate vectors span $\mathbb{C}^4$, some coordinate vector has nonzero projection into it. Selecting and normalizing such a projection gives the next orthonormal vector. The final two vectors form K. □

Apply Lemma 14 with $p = (\bar{b}_j)_j$ and with $p = (c_i)_i$. Throughout the seed ball keep the resulting frames fixed and write

$$G_B = B_0^*(B_0B_0^*)^{-1}, \qquad G_C = C_0(C_0^*C_0)^{-1}, \qquad \operatorname{ran} K_B = \ker B_0, \quad \operatorname{ran} K_C = \ker C_0^*. \tag{24}$$

Here $K_B^*K_B = K_C^*K_C = I_2$. Put $m_B = \sigma_{\min}(B_0)$, $m_C = \sigma_{\min}(C_0)$, and $m = \min(m_B, m_C) > 0$, so that $\|G_B\|_{\mathrm{op}} = 1/m_B$, $\|G_C\|_{\mathrm{op}} = 1/m_C$. Fix the orthogonal projections

$$P_K = K_BK_B^*, \qquad P_B = I - P_K = G_BB_0, \qquad P_C = I - K_CK_C^* = G_CC_0^*. \tag{25}$$

Also

$$A_0 = \|E_0\|_{\mathrm{op}} = \sqrt{2 + |1 + a|}, \qquad A_* = \min(2, A_0 + R_s). \tag{26}$$

Both A_0 and $\|E'\|_{\mathrm{op}}$ are at most A_*: indeed $A_0 \le 2$, every entry of E' has modulus one so $\|E'\|_{\mathrm{op}} \le \|E'\|_F = 2$, and $\|E'\|_{\mathrm{op}} \le A_0 + e_1 \le A_0 + R_s$.

The numerical construction of B_0, C_0 fixes particular coordinate choices for K_B, K_C. Acceptance checks on rank and residual norms certify that these choices are well defined. A failed frame check leaves the seed unresolved and never excludes it.

A unitary associated to every feasible completion.

The polar decomposition associates a unitary with any nonsingular square matrix, and its nearest-unitary property in unitarily invariant norms is classical: Theorem 1 of Fan and Hoffman, and Theorem 1.1 with Corollary 2.3 of Higham's 1986 study. Here the compressed completion has an explicit positive defect.

Lemma 15 (Polar comparison)**.** *For a feasible completion let $Y = K_C^*D'K_B$. Then*

$$6I - Y^*Y = K_B^*B'^*B'K_B + K_B^*D'^*P_CD'K_B \succeq 0, \tag{27}$$

$$\operatorname{tr}(6I - Y^*Y) \le (b_1')^2 + \frac{(\|E'\|_{\mathrm{op}}b_1' + \sqrt{6}\,c_1')^2}{m_C^2} \le q, \tag{28}$$

where

$$q = R_s^2\Big(1 + \frac{A_*^2 + 6}{m_C^2}\Big). \tag{29}$$

If $q < 6$ the polar factor $V = Y(Y^*Y)^{-1/2}$ lies in $U(2)$ and

$$\|Y - \sqrt{6}V\|_F \le p := \frac{q}{\sqrt{6} + \sqrt{6 - q}}. \tag{30}$$

Proof. The lower-right block of $H'^*H' = 6I$ is $B'^*B' + D'^*D' = 6I$. Compressing by K_B and subtracting $Y^*Y = K_B^* D'^*(I - P_C)D'K_B$ proves (27). Its trace is $\|B'K_B\|_F^2 + \|P_C D'K_B\|_F^2$. Since $B_0 K_B = 0$ the first term is at most $(b_1')^2$. Off-diagonal column orthogonality gives $C'^*D' = -E'^*B'$, hence $C_0^* D'K_B = -E'^*\Delta B K_B - \Delta C^* D'K_B$. Now $P_C = G_C C_0^*$ and $\|D'\|_{\mathrm{op}} \le \sqrt{6}$, giving the first inequality in (28). Cauchy–Schwarz with (22) gives the second.

Let ℓ_1, ℓ_2 be the eigenvalues of $6I - Y^*Y$. They are nonnegative with sum at most $q < 6$, so Y has positive singular values $\sqrt{6 - \ell_j}$ and its polar factor is unitary. Moreover

$$\sqrt{6} - \sqrt{6 - \ell_j} = \frac{\ell_j}{\sqrt{6} + \sqrt{6 - \ell_j}} \le \frac{\ell_j}{\sqrt{6} + \sqrt{6 - q}}.$$

Taking the Euclidean norm of these two differences and using $\sqrt{\ell_1^2 + \ell_2^2} \le \ell_1 + \ell_2 \le q$ proves (30). □

The inequality $q < 6$ is checked using outward bounds; in the specified parameter range the stronger $q < .32$ follows from the accepted frame checks and the seed radius. Hence every feasible completion of an accepted seed ball is assigned at least one $V \in U(2)$.

The unitary cells and their Taylor bounds.

Lemma 16 (A closed covering of $U(2)$)**.** *Every $V \in U(2)$ has a representation*

$$V = e(\alpha) \begin{pmatrix} e(\beta)\cos(2\pi t) & e(\gamma)\sin(2\pi t) \\ -e(-\gamma)\sin(2\pi t) & e(-\beta)\cos(2\pi t) \end{pmatrix}, \tag{31}$$

with $0 \le \alpha \le \frac{1}{2}$, $0 \le t \le \frac{1}{4}$, $0 \le \beta, \gamma \le 1$. For any integer N_u divisible by four, the products of closed intervals of width $1/N_u$ in these ranges cover the parameter domain, with bin counts $N_u/2, N_u/4, N_u, N_u$.

Proof. Choose a square root of $\det V$ having phase $\alpha \in [0, 1/2]$ in cycles. Then $e(-\alpha)V \in SU(2)$ has the form $\begin{pmatrix} z & w \\ -\overline{w} & \overline{z} \end{pmatrix}$ with $|z|^2 + |w|^2 = 1$. Choose $t \in [0, 1/4]$ with $|z| = \cos(2\pi t)$, $|w| = \sin(2\pi t)$, and phases β, γ for z, w; a zero entry may have arbitrary phase. So no boundary case requires division by an entry, and the closed interval products cover the domain directly. □

Fix one cell and denote its midpoint matrix by U. If V is represented inside the cell, let

$$\theta = (\theta_\alpha, \theta_t, \theta_\beta, \theta_\gamma), \qquad |\theta_j| \le h := \pi/N_u \tag{32}$$

be its displacement from the midpoint in radians. Denote by $dU[\theta]$ the midpoint angular differential applied to θ. All four derivatives are with respect to radian angles, even though (31) uses cycle coordinates.

Lemma 17 (Uniform cell Taylor estimates)**.** *For the cell and displacement in (32),*

$$\|V - U\|_F \le \sqrt{6}\, h, \qquad \|V - U - dU[\theta]\|_F \le \frac{\sqrt{42}}{2}\, h^2. \tag{33}$$

Proof. Follow the straight segment in the four angle coordinates, writing its matrix as $U(\tau)$, $0 \le \tau \le 1$. Abbreviate the four constant angular velocities as $x = \theta_\alpha$, $y = \theta_t$, $z = \theta_\beta$, $w = \theta_\gamma$, and let T be the current radian t angle. Differentiating the four entries yields

$$\|U'(\tau)\|_F^2 = 2(x^2 + y^2 + z^2\cos^2 T + w^2 \sin^2 T) \le 6h^2.$$

For the two diagonal entries, the squared magnitudes of the real factors in the second derivative have sum

$$\begin{aligned}&(((x+z)^2+y^2)^2+((x-z)^2+y^2)^2)\cos^2 T\\&\quad=2\big((x^2+z^2+y^2)^2+4x^2z^2\big)\cos^2 T\le 26h^4\cos^2 T,\end{aligned}$$

and the imaginary factors contribute $4y^2((x+z)^2+(x-z)^2)\sin^2 T\le 16h^4\sin^2 T$. The off-diagonal entries give the same bounds with z replaced by w and cosine and sine interchanged. Real and imaginary factors add in squared magnitude, so $\|U''(\tau)\|_F^2\le 42h^4$. Integrating the first derivative and then using the integral Taylor remainder with weight $1-\tau$ proves (33). □

Concretely, if U_{ij} are the entries of U, the alpha derivative is iU; the beta derivative has diagonal entries $iU_{00}, -iU_{11}$ and zero off-diagonal entries; the gamma derivative has off-diagonal entries $iU_{01}, -iU_{10}$ and zero diagonal; the t derivative differentiates the displayed sine and cosine factors. These are the four angular derivatives assembled by `uc::init` and `uc::initDeriv`.

Comparison with a cell center.

For the fixed seed and unitary cell define

$$D=D_{\mathrm{span}}+D_{\mathrm{ker}},\qquad D_{\mathrm{span}}=-C_0E_0^*G_B^*,\qquad D_{\mathrm{ker}}=\sqrt{6}\,K_CUK_B^*,\qquad Z=D'-D.\tag{34}$$

The symbol D throughout denotes this comparison block, which need not be entrywise flat; it is not asserted to be the lower-right block of any Hadamard matrix at the seed center.

Lemma 18 (Orthogonal comparison pieces)**.** *The matrices ZP_B, P_CZP_K, and $(I-P_C)ZP_K$ are pairwise Frobenius-orthogonal and sum to Z. They satisfy*

$$ZP_B=(-\Delta CE_0^*-C'\Delta E^*-D'\Delta B^*)G_B^*,\tag{35}$$

$$P_CZP_K=G_C(-E'^*\Delta B-\Delta C^*D')P_K,\tag{36}$$

$$(I-P_C)ZP_K=K_C(Y-\sqrt{6}\,U)K_B^*.\tag{37}$$

Consequently every feasible completion associated to this unitary cell satisfies

$$\|Z\|_F\le z:=\sqrt{\left(\frac{R_s(\sqrt{6}+A_*)}{m}\right)^2+(6h+p)^2}\,.\tag{38}$$

Proof. The right projections P_B, P_K are complementary and orthogonal, and within the right P_K-part the left projections $P_C, I-P_C$ are complementary and orthogonal; this proves the decomposition and the orthogonality. Row orthogonality gives $D'B'^*=-C'E'^*$. Since $P_B=B_0^*G_B^*$, $D_{\mathrm{ker}}P_B=0$, and $D_{\mathrm{span}}P_B=D_{\mathrm{span}}$, subtracting the comparison block produces (35). The identity $C'^*D'=-E'^*B'$, together with $B_0P_K=0$ and $P_CDP_K=0$, gives (36). The definitions of Y, K_B, K_C give (37).

Put $r_{EB}=\sqrt{e_1^2+(b_1')^2}$. The four lower rows of H' are orthogonal and have norm $\sqrt{6}$, so $\|[C'\ D']\|_{\mathrm{op}}=\sqrt{6}$. Therefore

$$\|ZP_B\|_F\le\frac{\sqrt{6}\,r_{EB}+A_0c_1'}{m_B},$$

$$\|P_CZP_K\|_F\le\frac{A_*b_1'+\sqrt{6}\,c_1'}{m_C}\le\frac{A_*r_{EB}+\sqrt{6}\,c_1'}{m}.$$

The vector of these two norms is bounded componentwise by

$$\frac{1}{m}\begin{pmatrix}\sqrt{6} & A_* \\ A_* & \sqrt{6}\end{pmatrix}\begin{pmatrix} r_{EB} \\ c_1'\end{pmatrix}.$$

The displayed matrix has operator norm $\sqrt{6}+A_*$, and its input norm is at most R_s by (22). For the third piece, multiplication by K_C, K_B preserves Frobenius norm, so

$$\|Y-\sqrt{6}\,U\|_F \le \|Y-\sqrt{6}\,V\|_F + \sqrt{6}\,\|V-U\|_F \le p+6h.$$

Pythagoras for the three pieces gives (38). □

Lemma 19 (A row-block operator bound)**.** *For every cell midpoint $U \in U(2)$,*

$$\|[C_0\ D]\|_{\mathrm{op}} \le L := \max\Big(\sqrt{6},\ \|C_0\|_{\mathrm{op}}\sqrt{1+(A_0/m_B)^2}\Big). \tag{39}$$

Also $\|DP_K\|_{\mathrm{op}} = \sqrt{6}$ and

$$\|D\|_{\mathrm{op}} \le \max\Big(\sqrt{6},\ \frac{\|C_0\|_{\mathrm{op}}A_0}{m_B}\Big). \tag{40}$$

Proof. The two terms $D_{\mathrm{span}}, D_{\mathrm{ker}}$ have orthogonal right domains and orthogonal ranges, so the cross terms in DD^* vanish. Using $G_B^*G_B = (B_0B_0^*)^{-1}$ gives

$$[C_0\ D][C_0\ D]^* = C_0\big[I + E_0^*(B_0B_0^*)^{-1}E_0\big]C_0^* + 6K_CK_C^*. \tag{41}$$

The first summand has range in $\operatorname{ran} C_0$ and operator norm at most $\|C_0\|_{\mathrm{op}}^2(1+A_0^2/m_B^2)$; the second is six times the projection onto the orthogonal complement. The total norm is the larger of these two, giving (39). Omitting the $C_0C_0^*$ contribution gives (40). Finally $DP_K = D_{\mathrm{ker}}$, whose two nonzero singular values are $\sqrt{6}$. □

A uniform linearization remainder.

Let E_1, B_1, C_1 be the seed first variations: at a varied entry, multiply the center entry by i times the corresponding component of s; at fixed entries the variation is zero. Define the linear map

$$\begin{aligned}\mathcal{J}(s,\theta) = {} & (-C_1E_0^* - C_0E_1^* - DB_1^*)G_B^* \\ & + G_C(-E_0^*B_1 - C_1^*D)P_K + \sqrt{6}\,K_C\,dU[\theta]K_B^*.\end{aligned} \tag{42}$$

This is a necessary linear comparison inferred from the exact orthogonality equations. It is not obtained by assuming frames vary differentiably with the seed, and it does not assume the existence of an exact completion at every seed center.

Lemma 20 (Linear remainder bound)**.** *Every feasible completion associated to the cell satisfies*

$$\|Z-\mathcal{J}(s,\theta)\|_F \le E_r, \qquad E_r = \sqrt{e_{\mathrm{off}}^2 + \left(\frac{\sqrt{6\cdot 42}}{2}\,h^2 + p\right)^2}, \tag{43}$$

where

$$e_{\mathrm{off}} = \frac{R_s^2(L+A_0+1)}{2m} + \frac{zR_s}{m}. \tag{44}$$

Proof. The first two terms of (42) lie in the spaces of the first two projections of Lemma 18; the last term lies in the third. Bound the remainders in each.

First replace C', D' in (35) by C_0, D. The additional numerator terms are $-\Delta C\,\Delta E^* - Z\Delta B^*$. Replacing E', D' in (36) by E_0, D leaves $-\Delta E^*\Delta B - \Delta C^* Z$. Across the two orthogonal output spaces the seed-only product terms have combined Frobenius norm at most

$$\frac{e_1\sqrt{(b_1')^2 + (c_1')^2}}{m} \le \frac{R_s^2}{2m}.$$

The product terms involving Z have combined norm at most

$$\frac{\|Z\|_F\sqrt{(b_1')^2 + (c_1')^2}}{m} \le \frac{zR_s}{m}.$$

These use submultiplicativity with a Frobenius norm on the varying factor, the bounds on G_B, G_C, and contraction by P_K.

It remains in the first two spaces to replace each of $\Delta E, \Delta B, \Delta C$ by its first variation. Since $|e^{ix} - 1 - ix| \le x^2/2$, the combined Frobenius norm of the three Taylor remainders is at most

$$\frac{1}{2}\Big(\sum_{j=1}^{9} s_j^4\Big)^{1/2} \le \frac{1}{2}\|s\|^2 \le \frac{1}{2}R_s^2.$$

For arbitrary remainder inputs F_E, F_B, F_C put $x = (\|F_E\|_F^2 + \|F_B\|_F^2)^{1/2}$ and $y = \|F_C\|_F$. Lemma 19 bounds the norms of their two linear output pieces componentwise by

$$\frac{1}{m}\begin{pmatrix} L & A_0 \\ A_0 & \sqrt{6} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

For the second component use $\|DP_K\|_{\mathrm{op}} = \sqrt{6}$. Since $L \ge \sqrt{6}$ the matrix has operator norm at most $L + A_0$. The combined seed Taylor contribution is therefore at most $R_s^2(L + A_0)/(2m)$, and adding the two product bounds gives (44).

In the third space the exact remainder is $K_C[Y - \sqrt{6}\,U - \sqrt{6}\,dU[\theta]]K_B^*$. Insert $\sqrt{6}\,V$ and apply the polar and unitary Taylor bounds: its norm is at most $p + \sqrt{6\cdot 42}\,h^2/2$. This space is orthogonal to the first two, so Pythagoras proves (43). □

The entrywise form for assembling (42) is: for $0 \le i, j, k < 4$ the shared seed coefficient is

$$-c_i\overline{ia(G_B)_{j1}},$$

the b_k coefficient is

$$-\big[(G_C)_{i0} + (G_C)_{i1}\overline{a}\big]\, ib_k(P_K)_{kj} - D_{ik}\overline{ib_k(G_B)_{j1}},$$

and the c_k coefficient is

$$-\mathbf{1}_{i=k}\, ic_k\big(\overline{(G_B)_{j0}} + \overline{a(G_B)_{j1}}\big) - (G_C)_{i1}\overline{ic_k}(D_{\ker})_{kj}.$$

For the angular coefficients, if the entries of U are flattened in row-major order, use

$$W_{ij,l} = \sqrt{6}\,(K_C)_{i,\lfloor l/2\rfloor}\overline{(K_B)_{j,l \bmod 2}}, \qquad 0 \le l < 4,$$

and contract with each angular derivative of U. These are the fixed and moving parts of `block::setup`, `makeD`, and `makeJ`.

Necessary flatness and separating inequalities.

We have bounded the total block displacement $Z = D' - D$ by z and its remainder after the linear comparison $\mathcal{J}(s,\theta)$ by E_r. The next step uses these bounds inside the necessary equations $|D'_{ij}| = 1$.

Lemma 21 (Norm exclusion)**.** *A feasible completion associated to a cell must satisfy*

$$\sum_{i,j=0}^{3} (|D_{ij}| - 1)^2 \le z^2. \tag{45}$$

Proof. Every D'_{ij} has modulus one, and the reverse triangle inequality gives $\big||D_{ij}| - 1\big| \le |D'_{ij} - D_{ij}|$. Sum squares and use (38). □

For the second test define two complex constants and their linear parts by

$$\begin{aligned} c_X &= (1 + a + \textstyle\sum_j b_j)/i, & J_X s &= a s_a + \textstyle\sum_j b_j s_{b_j}, \\ c_Y &= (1 + a + \textstyle\sum_i c_i)/i, & J_Y s &= a s_a + \textstyle\sum_i c_i s_{c_i}. \end{aligned}$$

The corresponding exact row and column sums vanish, so Taylor's formula gives $c_X + J_X s + \rho_X = 0$ and $c_Y + J_Y s + \rho_Y = 0$ with

$$|\rho_X| \le \tfrac{1}{2} R_s^2, \qquad |\rho_Y| \le \tfrac{1}{2} R_s^2. \tag{46}$$

Take real and imaginary parts of these two equations and append the following sixteen real constants and linear forms:

$$c_{ij} = \frac{|D_{ij}|^2 - 1}{2}, \qquad J_{ij}(s,\theta) = \mathrm{Re}\big(\overline{D_{ij}}\, \mathcal{J}_{ij}(s,\theta)\big). \tag{47}$$

Denote the resulting real constant vector by $c_{\mathrm{all}} \in \mathbb{R}^{20}$ and the real linear matrix by $J_{\mathrm{all}} \in \mathbb{R}^{20\times 13}$, with the first nine columns corresponding to s and the last four to θ.

Lemma 22 (Directional exclusion)**.** *Let $y \in \mathbb{R}^{20}$ be arbitrary, split as two pairs $y_X, y_Y \in \mathbb{R}^2$ and a flat part $y_D \in \mathbb{R}^{16}$, and write $g = J_{\mathrm{all}}^* y = (g_s, g_\theta) \in \mathbb{R}^9 \times \mathbb{R}^4$. Every feasible completion of the cell satisfies*

$$\begin{aligned} y \cdot c_{\mathrm{all}} \le{}& R_s \|g_s\| + h \|g_\theta\|_1 + E_r \|D \circ y_D\|_F \\ &+ \frac{1}{2} z^2 \max\big(0, \max_{i,j}(-y_{D,ij})\big) + \frac{1}{2} R_s^2 (\|y_X\| + \|y_Y\|). \end{aligned} \tag{48}$$

Here $\circ$ is entrywise multiplication, with the sixteen flat coordinates arranged as a 4×4 matrix.

Proof. Write $R = Z - \mathcal{J}(s,\theta)$, so $\|R\|_F \le E_r$. Expanding the exact flatness equation gives

$$0 = \frac{|D'_{ij}|^2 - 1}{2} = c_{ij} + J_{ij}(s,\theta) + \mathrm{Re}\big(\overline{D_{ij}} R_{ij}\big) + \frac{1}{2}|Z_{ij}|^2.$$

Weight by y_D, and similarly weight the two complex seed equations by y_X, y_Y through their real coordinate pairs. Move every term other than $y \cdot c_{\mathrm{all}}$ to the right. The linear contribution is bounded by $R_s\|g_s\| + h\|g_\theta\|_1$ using the seed ball and angle box; real Cauchy–Schwarz bounds the flat remainder contribution by $E_r\|D \circ y_D\|_F$; the seed remainders contribute the last term by (46).

The quadratic term has a fixed sign before weighting. After it is moved to the right its contribution is

$$-\frac{1}{2} \sum_{i,j} y_{D,ij} |Z_{ij}|^2 \le \frac{1}{2} \max\big(0, \max_{i,j}(-y_{D,ij})\big) \sum_{i,j} |Z_{ij}|^2.$$

Use $\|Z\|_F \le z$. This proves the remaining term and the stated inequality. □

Corollary 23 (Dual certificates)**.** *A cell contains no feasible completion if* (45) *fails, or if some real direction y violates* (48)*. The choice of y may be heuristic; only its final inequality must be certified. Positive normalization of y preserves the certificate.*

Proof. Both inequalities are necessary for every feasible completion in the cell. All terms of (48) are positively homogeneous in y, so a strict violation is a separating, or dual, certificate, independent of how the direction was found. □

The implementation stacks exactly these twenty equations in `eq`. The routine `trial::pass` proposes directions by regularized linear solves and weight updates; it excludes only when `trial::certificate` verifies the final inequality. That routine checks finiteness, divides by the largest absolute coordinate, and treats the resulting stored coefficients as the direction. It uses the negative part of the flat weights, as required by (48). No accuracy or convergence assertion about the proposing solves is needed. The used Stage A path is `Asearch::testA`; the auxiliary `Usearch::test` and `scalar` routines do not supply additional exclusion tests for this run. The actual norm and directional comparisons include the margins proved in the certified arithmetic.

Finite recursion and unresolved cells.

Proposition 24 (Completion coverage conditional on successful termination)**.** *Fix a seed ball whose frame and bound constructions are certified. Suppose every final unitary cell is either excluded by a certified necessary-condition test or covered by a certified phase ball, as in the certified arithmetic. Then those phase balls cover every feasible Hadamard completion of the seed ball, up to the equivalences of the phase-ball certificates. The Stage A recursion establishes exactly this if it terminates with zero unresolved cells.*

Proof. Given any feasible completion, Lemma 15 produces a unitary V. Lemma 16 places one of its parameter representations in a closed base cell. Every certified exclusion test is necessary for that completion, so it cannot exclude a cell containing that representation. If a cell is subdivided, the two closed children in each coordinate cover the parent interval; their sixteen products cover the parent cell. Hence the representation can be followed into at least one child. A successful phase-ball certificate covers every feasible completion associated to its entire cell, so it covers the chosen completion if that outcome is reached.

This is a finite induction through the processed cell tree. Zero unresolved cells means every branch ends in a certified exclusion or a certified phase-ball assignment; the branch followed by a feasible completion cannot end in exclusion and therefore ends in coverage. □

In the stated invocation the base grid is $N_u = 16$. Retained cells are split to $N_u = 32$ and then 64 by replacing every bin index b_j with both $2b_j$ and $2b_j + 1$. At the first level only the norm test is needed. At 64 and deeper, every unexcluded cell is offered to the phase-ball coverage routine; unsuccessful coverage causes all sixteen children to be processed with the same seed and the same fixed frames, using the new value of h. Subdivision is allowed through $N_u = 512$.

A coverage or snapping failure does not exclude a cell. A live cell at the depth cap increments the unresolved count. Exceeding the deeper-node cap of 1,000,000 sets the give-up flag and increments that count; later short returns under that flag cannot restore successful status. A failed frame construction is recorded as a failed seed. The driver records every seed with unresolved cells and returns failure if any seed failed. Thus a negative or global-coverage conclusion requires successful processing of all prescribed seed indices, including all shards, with no failed seed or unresolved cell. Retaining some valid phase balls from an unsuccessful seed does not certify the unprocessed remainder; these conditions are part of the computational obligation. They do not follow from the local inequalities alone.

Certified phase balls and their atlas.

This section converts a cell retained by into a ball of real phase matrices. For a center P, $e(P)$ need not be an exact Hadamard matrix; the certificates control the center's residual and the displacement of every exact solution inside the ball. Phase matrices and their ball radii are measured in cycles, except that angular displacements at the code level are measured in radians and stated explicitly. Real 6×6 matrices are identified with $\mathbb{R}^{36}$ in row-major order; a displacement is *horizontal* if it lies in $\operatorname{ran}\Pi$. All vector norms are Euclidean, phase matrix norms are Frobenius, and stars on real matrices mean transpose.

Residual and Root certificates.

For a real phase matrix P define $F(P) \in \mathbb{C}^{15}$ by

$$F_{ij}(P) = \frac{1}{2\pi i} \sum_{g=0}^{5} e(P_{ig} - P_{jg}), \qquad 0 \le j < i \le 5, \tag{49}$$

with edges ordered by increasing i, then increasing j; a complex coordinate is stored as real part then imaginary part. We also regard $F(P)$ as a real 30-vector. Since every entry of $e(P)$ has modulus one, $F(P) = 0$ is equivalent to the row Hadamard conditions, and hence to all of them. Its real derivative $L_P : \mathbb{R}^{36} \to \mathbb{R}^{30}$ in complex notation is

$$(L_P x)_{ij} = \sum_{g=0}^{5} e(P_{ig} - P_{jg})(x_{ig} - x_{jg}). \tag{50}$$

The factor $2\pi i$ from differentiation cancels the denominator in (49). These are the residual and full derivative in `buildH::gen`.

Definition 25 (Root certificate)**.** A `Root` consists of a stored real center P, a stored real $36 \times k$ matrix T, and nonnegative constants C, `n0`, `id`, `ort`, `hor`, `mnorm`, `froot`. The source stores column a of T as `T[a]` and uses $0 \le k \le 10$. Write

$$n_P(s) = \texttt{n0} + \texttt{id}\, s + C\, s^2, \qquad s \ge 0. \tag{51}$$

The data are certified if the basic bounds

$$\|T^*T - I\| \le \texttt{ort}, \qquad \|(I - \Pi)T\| \le \texttt{hor}, \qquad \|F(P)\| \le \texttt{froot} \tag{52}$$

hold and in addition, for every horizontal δ with $F(P + \delta) = 0$ and $\eta = T^*\delta$, $s = \|\delta\|$,

$$\|(I - TT^*)\delta\| \le n_P(s), \tag{53}$$

$$s^2 \le (1 + \texttt{ort})\|\eta\|^2 + \|(I - TT^*)\delta\|^2, \tag{54}$$

and, for every such δ and every horizontal v, with $h' = \delta - v$,

$$\|(I - TT^*)h'\| \le \big[\texttt{id} + C(\|\delta\| + \|v\|)\big]\|h'\| + \texttt{mnorm}\,\|F(P + v)\|. \tag{55}$$

These assertions have no radius restriction. A Root certificate neither requires $F(P) = 0$ nor asserts the existence of an exact solution. A radius is attached only when a phase ball is recorded.

For a displacement δ of an exact solution, given $s_0 \ge \|\delta\|$ and $t_0 \ge \|T^*\delta\|$, nonnegativity of the coefficients of n_P gives

$$\|\delta\| \le \big(n_P(s_0)^2 + (1 + \texttt{ort})t_0^2\big)^{1/2}.$$

This is used to improve a radius that already covers all exact completions in the cell. The difference bound is used to compare displacements from two centers during refinement.

Remark 26. The columns of T are numerical directions only. Neither exact horizontality nor exact orthogonality is assumed; in particular $\|T\| \le \sqrt{1+\texttt{ort}}$. The `hor` term appears explicitly when raw coordinate displacements are projected. The normal estimates bound the omitted directions without setting them to zero. No constant-rank or manifold hypothesis on the exact solution set is used.

Constructing Root certificates.

The construction uses an auxiliary real map $M : \mathbb{R}^{30} \to \mathbb{R}^{36}$ that turns the residual equation into a bound on $(I - TT^*)\delta$. This map is used only to establish the Root assertions and is not serialized.

Lemma 27 (Quadratic and difference remainders)**.** *There is a real* 30*-vector* $\mathcal{R}_P(x)$ *with* $F(P+x) = F(P) + L_P x + \mathcal{R}_P(x)$*. Put* $a_{ij}(x) = \|x_i - x_j\|^2$ *(where* x_i *is row* i *of* x*). Then*

$$|(\mathcal{R}_P(x))_{ij}| \le \pi a_{ij}(x), \tag{56}$$

$$|(\mathcal{R}_P(x) - \mathcal{R}_P(v))_{ij}| \le \pi\sqrt{a_{ij}(x-v)}\,(\sqrt{a_{ij}(x)} + \sqrt{a_{ij}(v)}). \tag{57}$$

For a real unit $y \in \mathbb{R}^{36}$ *let* $w_{ij}(y)$ *be the Euclidean norm of the two edge coordinates of* M^*y*. If*

$$\sum_{i>j} w_{ij}(y)\,a_{ij}(x) \le b\|x\|^2 \quad \text{for every horizontal } x \text{ and every unit } y, \tag{58}$$

then for horizontal x, v*,*

$$\|M\mathcal{R}_P(x)\| \le \pi b\|x\|^2, \tag{59}$$

$$\|M(\mathcal{R}_P(x) - \mathcal{R}_P(v))\| \le \pi b\,\|x - v\|(\|x\| + \|v\|). \tag{60}$$

Proof. For real t put $r(t) = (e(t) - 1 - 2\pi i t)/(2\pi i)$. Its derivative is $e(t) - 1$, of modulus at most $2\pi|t|$, so $|r(t)| \le \pi t^2$ by integrating from zero. On the segment from v to x, convexity of absolute value gives

$$|r(x) - r(v)| \le 2\pi|x - v| \int_0^1 |(1-u)v + ux|\,du \le \pi|x - v|(|x| + |v|).$$

Apply to each scalar row difference and sum over g. The unit factors $e(P_{ig} - P_{jg})$ do not affect modulus. Cauchy–Schwarz in g gives (57).

Pairing $M\mathcal{R}_P(x)$ with a real unit y and using the two-coordinate Cauchy–Schwarz inequality on each edge proves (59). For the difference, apply separately

$$\sum_{i>j} w_{ij}\sqrt{a_{ij}(x-v)a_{ij}(x)} \le \Big(\sum_{i>j} w_{ij}a_{ij}(x-v)\Big)^{1/2}\Big(\sum_{i>j} w_{ij}a_{ij}(x)\Big)^{1/2} \le b\|x-v\|\|x\|,$$

and the same inequality with v in the last factor. Take the supremum over y; the same constant controls both remainders. □

Lemma 28 (Two computable remainder constants)**.** *Two sufficient choices in* (58) *are* $b = \sqrt{8}\,\|M\|$ *and* $b = \sqrt{6}\,\|\mathcal{S}\|$*, where the* 30×216 *real matrix* $\mathcal{S}$ *has row*

$$\mathcal{S}_{a,:} = M^*_{:,a} \otimes (\mathbf{e}_i - \mathbf{e}_j)^*$$

when a *is either of the two real-coordinate indices of edge* $i > j$*. Consequently* π *times the smaller of any certified upper bounds for these two quantities is a valid* C*.*

Proof. For horizontal x with row vectors summing to zero, let $s = \|x\|$ and $u_i = \|x_i\|^2$. Expansion gives

$$\sum_{i>j} a_{ij}(x) = 6s^2,$$
$$\sum_{i>j} a_{ij}(x)^2 = 6\sum_i u_i^2 + s^4 + 2\sum_{i,j}(x_i \cdot x_j)^2. \tag{61}$$

The cross terms use $\sum_{j\neq i} x_i \cdot x_j = -u_i$. Also $u_i \le 5(s^2 - u_i)$ and $\sum_{i,j}(x_i \cdot x_j)^2 \le s^4$. Thus $\sum_i u_i^2 \le (5/6)s^4$, and the second sum in (61) is at most $8s^4$. Cauchy–Schwarz on the edges, with $\sum w_{ij}(y)^2 = \|M^* y\|^2$, proves the first choice of b. For the second, apply $\mathcal{S}$ to $y \otimes x_{:,g}$ and sum squared norms over g: for unit y this gives $\sum w_{ij}(y)^2 a_{ij}(x) \le \|\mathcal{S}\|^2 s^2$. Another Cauchy–Schwarz, now using $\sum a_{ij} = 6s^2$, proves the claim. The row Gram of $\mathcal{S}$ is computable without forming its 216 columns:

$$(\mathcal{S}\mathcal{S}^*)_{ab} = \big((\mathbf{e}_i - \mathbf{e}_j) \cdot (\mathbf{e}_{i'} - \mathbf{e}_{j'})\big)(M_{:,a} \cdot M_{:,b}), \tag{62}$$

with the edge labels repeated for real and imaginary parts. This is the sign-factor formula in `buildH::patch`. □

Proposition 29 (Sufficient conditions for a Root certificate)**.** *Let P, T and the nonnegative constants be as in Definition 25. Suppose (52) holds and an auxiliary stored real 36×30 matrix M satisfies*

$$\|M\| \le \texttt{mnorm}, \qquad \texttt{mnorm}\,\texttt{froot} \le \texttt{n0}, \qquad \|\Pi - TT^* - ML_P\Pi\| \le \texttt{id}, \tag{63}$$

and C is an upper bound for a remainder constant πb in Lemma 27. Then these data form a certified Root.

Proof. Let δ be horizontal with $F(P+\delta) = 0$ and put $\eta = T^*\delta$. Write $E = \Pi - TT^* - ML_P\Pi$. Taylor's formula at the exact solution gives

$$(I - TT^*)\delta = E\delta - MF(P) - M\mathcal{R}_P(\delta).$$

The certificate and Lemma 27 prove (53). For the squared inequality (no projector assumption on TT^*),

$$\|\delta - T\eta\|^2 = \|\delta\|^2 - 2\|\eta\|^2 + \eta^*(T^*T)\eta \ge \|\delta\|^2 - (1 + \texttt{ort})\|\eta\|^2;$$

this holds for arbitrary δ. For horizontal v, put $h' = \delta - v$ and subtract Taylor formulas at δ and v. Since h' is horizontal and $F(P+\delta) = 0$,

$$(I - TT^*)h' = Eh' - MF(P+v) - M\big(\mathcal{R}_P(\delta) - \mathcal{R}_P(v)\big).$$

Apply (60). This proves (55), including the residual term at the possibly inexact $P + v$. □

The combination of a numerical factorization with a checked Gram defect is standard in verified matrix-norm estimation, as in Rump's 2011 study of verified singular-value bounds. The bounds come from the displayed residual identities and the certified arithmetic estimates, including the error in the proposed tangent frame.

Proposition 30 (Certificates from the direct checks)**.** *A successful call to `buildH::patch` under the arithmetic conditions of the certified arithmetic certifies (52), (63), and a valid remainder constant C in the sense of Definition 25.*

Proof. The direct checks, not the accuracy of the proposed eigendecomposition, establish the assertion. An approximate horizontal basis is proposed together with a splitting of the 25-coordinate horizontal Jacobian. Eigenvalues below .64 propose columns of T; the rest propose factors H_V, L_V for M. Every accepted factor entry has magnitude at most 2, and at most 25 columns occur. More than 10 proposed T columns causes failure.

Certified absolute row sums of the column Grams of H_V, L_V bound their squared operator norms. The bound `mnorm` also includes the difference between the accumulated stored M and the exact product $H_V L_V^*$. The required additions and the $2.1 \cdot 10^{-11}$ Frobenius bound on this difference are proved in the certified arithmetic. Thus `mnorm` bounds the actual stored map, and the first choice in Lemma 28 supplies a certified C.

For the optional tensor improvement the row Gram (62) is formed, and a real change of basis U is proposed with a checked upper β for $\|U^*U - I\|$. If $\beta < .1$ then $\|U^{-1}\|^2 \le 1/(1-\beta)$. A certified absolute row sum bound r for $U^*(\mathcal{S}\mathcal{S}^*)U$ gives $\|\mathcal{S}\|^2 \le r/(1-\beta)$. This uses congruence by an invertible matrix, not assumed orthogonal diagonalization. The second choice in Lemma 28, with the one-sided margins certified alongside it, gives the alternative C. Both are upper bounds, so their minimum is allowed.

Finally the full 36-coordinate identity in (63) is evaluated, not just its proposed horizontal form. The padded Frobenius residual gives `id`. Corresponding checks on $T^*T - I$, $(I - \Pi)T$ and $F(P)$ give the basic bounds, and the stored `n0` is an upper bound for `mnorm froot`. The certified arithmetic proves the one-sided error estimates for all these checks and records their acceptance thresholds. Replacing operator norms by Frobenius norms only enlarges the bounds, so Proposition 29 applies to every successful call. □

Transport and alignment.

Callbacks in `variants` select a real 6×6 signed permutation matrix Φ using row and column entries of $e(P)$. The columns of Φ are the standard basis in a permuted order, and its entries lie in $\{0, \pm 1\}$. Thus $\Phi^*\Phi = \Phi\Phi^* = I$, and because Φ permutes real coordinates it induces on phase matrices a signed permutation preserving inner products, horizontality, and solution status. In particular, for a real P and its transported center $V = \Phi P$,

$$F(V) = F(\Phi P), \qquad L_{\Phi P}(\Phi\delta) = L_P\delta, \qquad F(V + \Phi\delta) = F(P + \delta) = 0. \tag{64}$$

Hence a Root transported by Φ remains a valid Root certificate with the same constants; the source writes such a transport through `Root::transport`. This is the map $P \mapsto \Phi P$ used below.

After transport, replace P by its dephasing $D_0(P)$. Dephasing is a row and column rephasing; by Lemma 5 this preserves extendibility, and it preserves F, L_P, and T up to the same rephasing.

For a horizontal displacement δ of an exact solution, write $\delta = \Delta + \Pi u/(2\pi) + \epsilon_X$ with the further split of u into a seed part s and an angular part d; then take a compact sub-ball.

Lemma 31 (Alignment)**.** *For real centers P, Q and integer matrix N,*

$$d = \Pi(D_0(P) - D_0(Q) - N) \quad \Longrightarrow \quad e(Q + d) \text{ is phase-equivalent to } e(P). \tag{65}$$

Under that alignment every exact solution displacement δ in a completion cell takes the form

$$\delta = \Delta + \Pi\Big(\epsilon + \frac{u}{2\pi}\Big), \qquad \Delta = \Pi(P_0 - D_0(P) - N), \tag{66}$$

with the same equivalence of δ-parameterized matrices.

Proof. For any real matrix X, $D_0(X) - X$ is constant on rows plus constant on columns, so $\Pi(D_0(X) - X) = \Pi X - \Pi X = 0$ after Π is applied. Thus $\Pi D_0(X) = \Pi X$, and $\Pi(D_0(P) - D_0(Q) - N) = \Pi(P - Q - N)$. The difference $P - N - (Q + d)$ then lies in $\ker \Pi$ plus integer entries, so it is a row and column rephasing. For every horizontal δ, the same fixed alignment acts on $Q + d + \delta$ to represent $P + \delta$ modulo the same rephasing. That is exactly extendibility preservation by Lemma 5. □

The standard `dephase` and `wrap` operations produce the integer choices in Δ; the exact Δ used in the arguments below is the ideal one, and bounds the computed vs. ideal difference.

Displacement bounds for a retained cell.

Fix a retained cell. Its seed displacement satisfies $\|s\| \le R_s$, its angular box satisfies $|\theta_a| \le h$, $1 \le a \le 4$, and $Z = D' - D$ has $\|Z\|_F \le z$ and $\|Z - \mathcal{J}(s, \theta)\|_F \le E_r$ as in . Let $\mathcal{I}_s = (7, 8, 9, 10, 11, 13, 19, 25, 31)$ denote the nine seed indices, and $\mathcal{I}_D$ the sixteen lower-right indices (i, j) with $2 \le i, j \le 5$. The first row and column have zero raw displacement.

For a retained cell we want a bound on the exact horizontal δ in (66). Let β be a fixed parameter; the actual codes use $\beta = .45$ or $\beta = .8$.

Lemma 32 (Arc and weighted-arc bounds)**.** *Suppose $m_j = |D_j| > 0$ for every lower-right position and put $D_j^\circ = D_j/m_j$. Choose the shortest angular displacement $d_j \in [-\pi, \pi]$ from D_j° to D_j'. Let* `slack` *be a nonnegative upper bound for $z^2 - \sum_j (m_j - 1)^2$, let* `mi` *be a positive lower bound for $\min_j m_j$, and choose $q_d \ge$* `slack`*/*`mi`*, $q_d < 3.8$. Then valid upper bounds are*

$$\|d\| \le \texttt{arc}, \qquad \Big(\sum_j m_j d_j^2\Big)^{1/2} \le \texttt{weighted}, \tag{67}$$

provided `arc` $\ge \sqrt{q_d}/\sqrt{1 - q_d/4}$ *and* `weighted` $\ge \sqrt{\texttt{slack}}/\sqrt{1 - q_d/4}$.

Proof. Since D_j' has modulus one,

$$|D_j' - D_j|^2 = (m_j - 1)^2 + m_j |D_j' - D_j^\circ|^2. \tag{68}$$

Summing proves $\sum_j m_j |D_j' - D_j^\circ|^2 \le$ `slack`, and hence $\sum_j |D_j' - D_j^\circ|^2 \le q_d$. In particular no individual chord reaches 2. If $c_j = |D_j' - D_j^\circ|$, $|d_j| = 2 \arcsin(c_j/2)$. On $0 \le c \le \sqrt{q_d}$ its derivative is at most $(1 - q_d/4)^{-1/2}$. Integrating from zero gives $|d_j| \le c_j (1 - q_d/4)^{-1/2}$; the two summed bounds follow. □

The source's `Candidate::prep` only attempts this argument when its direction and magnitude checks pass. It uses the stored flat `sum` and sets

$$\texttt{slack} = \max(0, z^2 - \texttt{sum} + 10^{-5}) + 10^{-10}, \qquad q_d = \texttt{slack}/\texttt{mi} + 10^{-8}.$$

The subtraction 10^{-6} from each computed magnitude gives `mi` and the upper reciprocal $\texttt{invmag}_j = 1/(m_j - 10^{-6}) + 10^{-9}$. The ratio has 10^{-7} added and the final arc quantities each have 10^{-8} added. shows these numbers have the one-sided properties of Lemma 32 for the ideal comparison block. Failure of any check means failure to cover this cell by this route, not an exclusion of the cell.

Lemma 33 (Stored center and support representation)**.** *Let P_0 be the stored* `Candidate` *phase matrix. On a successful preparation path there is a correction ϵ, in cycles, with*

$$\|\epsilon\| \le 2 \cdot 10^{-6}, \tag{69}$$

such that, up to entrywise integer choices, the actual feasible phase matrix has the form

$$P_0 + \epsilon + \frac{u}{2\pi}. \tag{70}$$

Here the raw angular matrix u is s at $\mathcal{I}_s$, is d at $\mathcal{I}_D$, and is zero elsewhere, so in particular

$$\|u\| \le \sqrt{R_s^2 + \mathtt{arc}^2}. \tag{71}$$

Let $X : \mathbb{R}^{13} \to \mathbb{R}^{36}$ give s at the seed positions and, at the lower-right positions, give

$$\big(X(s,\theta)\big)_j = \operatorname{Im}(\overline{D_j^\circ}\,\mathcal{J}(s,\theta)_j). \tag{72}$$

Then

$$u = X(s,\theta) + e_D + e_{\sin}, \qquad \|e_D\| \le E_r, \qquad \|e_{\sin}\|_1 \le \mathtt{arc}^3/6, \tag{73}$$

where both errors are supported on $\mathcal{I}_D$.

Proof. Only the lower-right block D' of $e(P_0)$ differs from D in something other than the seed displacement. The exact flatness equation $|D'_{ij}| = 1$ forces $D'_j = m_j e(d_j)$ up to a scalar. From Lemma 20,

$$D'_j = D_j + \mathcal{J}(s,\theta)_j + R_j, \qquad \|R\| \le E_r,$$

and Lemma 32 bounds $|d_j| \le \mathtt{arc}$. Divide D'_j by m_j: the real part is $\cos d_j$, the imaginary part is $\sin d_j$. Thus

$$\operatorname{Im}(\overline{D_j^\circ}\,\mathcal{J}(s,\theta)_j) = m_j^{-1}\operatorname{Im}(\overline{D_j^\circ}(D'_j - D_j)) + O(E_r) = d_j + O(E_r) + O(\mathtt{arc}^3).$$

The $O(E_r)$ contribution is at most E_r per component and the $O(\mathtt{arc}^3)$ sine expansion error is at most $\mathtt{arc}^3/6$ in one-norm, as asserted. The seed and D contributions add nothing further: the seed coordinates in u are exactly s, and the identity $\|u\|^2 = \|s\|^2 + \|d\|^2$ gives (71). The stored center P_0 differs from an exact solution by $\epsilon + u/(2\pi)$ up to integer entries. Dephasing an exact phase matrix changes P_0 by less than $2 \cdot 10^{-6}$ cycles per entry, as proved in the certified arithmetic. □

Lemma 34 (A direct displacement bound)**.** *Either of the following is an upper bound for $\Delta \cdot u$:*

$$S_1 = R_s\|(X^*\Delta)_s\| + h\|(X^*\Delta)_\theta\|_1 + E_r\|\Delta_D\| + \|\Delta_D\|_\infty \mathtt{arc}^3/6, \tag{74}$$

$$S_2 = R_s\|\Delta_s\| + \mathtt{weighted}\Big(\sum_{j\in\mathcal{I}_D} \Delta_j^2/m_j\Big)^{1/2}. \tag{75}$$

With $S_\Delta = \min(S_1, S_2)$, a valid upper bound is

$$\|\delta\| \le \Big(\|\Delta\|^2 + \frac{R_s^2 + \mathtt{arc}^2}{(2\pi)^2} + \frac{S_\Delta}{\pi}\Big)^{1/2} + \|\epsilon\|. \tag{76}$$

Proof. Equation (73), the seed ball, and the angle box give (74). Direct weighted Cauchy–Schwarz using (67) gives (75). Since Δ is horizontal, $\Delta \cdot \Pi u = \Delta \cdot u$. Expand the squared norm of $\Delta + \Pi u/(2\pi)$, contract by Π, and add $\|\Pi\epsilon\| \le \|\epsilon\|$ by the triangle inequality. □

Ellipsoid tangent bounds.

Partition $X = (X_s, X_\theta)$, and let T_s, T_D be the rows of T restricted to those index sets. From the angle box, $2\pi T^*\Delta + hT^*X_\theta[-1,1]^4$ is the convex hull of the 16 offsets

$$U_\sigma = 2\pi T^*\Delta + h\sum_{a=1}^{4} \sigma_a T^* X_{\theta,a}, \qquad \sigma_a \in \{-1, 1\}. \tag{77}$$

Lemma 35 (Ellipsoid containment)**.** *For real linear maps A, B, vectors $\|s\| \le r$, $\|v\| \le q$, and $\beta > 0$, the sum $As + Bv$ belongs to the ellipsoid $\{G^{1/2}w : \|w\| \le 1\}$ whenever*

$$G \succeq (1+\beta)r^2 AA^* + (1+1/\beta)q^2 BB^*.$$

Proof. For each real test vector z, the support is at most $r\|A^*z\| + q\|B^*z\|$, which by the inequality $(x+y)^2 \le (1+\beta)x^2 + (1+1/\beta)y^2$ is at most $(z^*Gz)^{1/2}$. Diagonalize G: the vectors satisfying $|z \cdot x| \le (z^*Gz)^{1/2}$ for every z are exactly $G^{1/2}$ times the unit ball, including when G is singular. □

By Lemma 35 with $\beta = .45$, the sum of the seed and e_D contributions to T^*u lies in an ellipsoid whenever

$$G_{\text{lin}} \succeq 1.45R_s^2(T^*X_s)(T^*X_s)^* + (1+1/.45)E_r^2 T_D^* T_D. \tag{78}$$

The remaining sine error has norm at most $(\max_{j\in\mathcal{I}_D}\|T_{j,:}\|)\texttt{arc}^3/6$. There is also a bound using the raw seed and weighted angle balls with a single offset $U_0 = 2\pi T^*\Delta$:

$$G_{\text{dir}} \succeq 1.8R_s^2 T_s^* T_s + (1+1/.8)\texttt{weighted}^2 T_D^* \operatorname{diag}(1/m_j) T_D. \tag{79}$$

Lemma 36 (Offset ellipsoid norm bound)**.** *Let $G \succeq 0$ and $U \in \mathbb{R}^k$. For every $\gamma > \lambda_{\max}(G)$ and $\|w\| \le 1$,*

$$\|U + G^{1/2}w\|^2 \le \gamma(1 + U^*(\gamma I - G)^{-1}U). \tag{80}$$

The triangle bound $\|U\| + \sqrt{\lambda_{\max}(G)}$ is also valid. For finitely many offsets ranging over their convex hull, the largest of these bounds is valid.

Proof. Put $A = \gamma I - G \succ 0$. Expanding and completing the square,

$$\|U + G^{1/2}w\|^2 \le \gamma + \|U\|^2 + 2w^*G^{1/2}U - w^*Aw \le \gamma + \|U\|^2 + U^*G^{1/2}A^{-1}G^{1/2}U.$$

Since G commutes with A^{-1}, the last two terms are $\gamma U^*A^{-1}U$. The triangle bound is immediate; convexity of norm gives the last claim. □

Let $\mathcal{E}(G; \{U_\sigma\})$ denote any certified upper bound from Lemma 36. Two valid tangent bounds are

$$\|T^*\delta\| \le \frac{\mathcal{E}(G_{\text{lin}}; \{U_\sigma\}) + \max_{j\in\mathcal{I}_D}\|T_{j,:}\|\,\texttt{arc}^3/6 + \texttt{hor}\sqrt{R_s^2 + \texttt{arc}^2}}{2\pi} + \sqrt{1+\texttt{ort}}\,\|\epsilon\|, \tag{81}$$

$$\|T^*\delta\| \le \frac{\mathcal{E}(G_{\text{dir}}; \{U_0\}) + \texttt{hor}\sqrt{R_s^2 + \texttt{arc}^2}}{2\pi} + \sqrt{1+\texttt{ort}}\,\|\epsilon\|. \tag{82}$$

Numerical realization. In `covercompute`, `off` is the computed $2\pi T^*\Delta$; `AB` is the computed T^*X; `U` holds the 16 corners. The covariance uses the stored `AB` coefficients and charges their ideal-coefficient error separately outside the ellipsoid. gives $\|\widehat{A} - T^*X\| < 1.5 \cdot 10^{-5}$, $\|(s,\theta)\| < .226$, so this substitution costs less than $3.4 \cdot 10^{-6}$ radians. For both covariance choices a diagonal $10^{-7}I$ is added. The inverses in the second choice are certified upper `invmag` values. proves PSD domination of the exact covariance expressions in the stored-coefficient model.

Bootstrap and local coverage.

Proposition 37 (Finite improvement of a certified radius)**.** *Suppose $s_0 \ge \|\delta\|$ and $t_0 \ge \|T^*\delta\|$ for every feasible solution in a completion cell, using the particular alignment (66). Starting from any such upper s, a further upper is*

$$\left(n_P(s)^2 + (1+\texttt{ort})t_0^2\right)^{1/2}. \tag{83}$$

Replacing s by this number only when it decreases s, and stopping after any finite number of steps, preserves coverage of every feasible solution in the cell.

Proof. Lemma 25, nonnegativity of the coefficients, and monotonicity of n_P give $\|\delta\|^2 \le n_P(\|\delta\|)^2 + (1 + \texttt{ort})\|T^*\delta\|^2 \le n_P(s)^2 + (1 + \texttt{ort})t_0^2$. The minimum of the two bounds is the current s and the right side, so the iterate never decreases below a true displacement. Finitely many strict improvements preserve the conclusion at the stopping point. □

Atlas merge of certified balls.

For a recorded phase ball, the `Atlas` stores the Root (P, T), the radius r, and the certified constants. Two entries are *merged* when a certified distance bound between their centers is at most the sum of their currently stored radii plus an allowance. The merged entry keeps the first Root and grows its radius to the validated sum.

Lemma 38 (Atlas merge)**.** *Suppose two Root records (P, r_P), (Q, r_Q) satisfy*

$$\|(P + \delta_P) - (Q + \delta_Q)\| \le d_{PQ} \quad \textit{whenever } \|\delta_P\| \le R_P,\ \|\delta_Q\| \le R_Q,\ F(P + \delta_P) = 0,$$

for a certified upper distance d_{PQ}. Then the union $\mathcal{S}(P, r_P) \cup \mathcal{S}(Q, r_Q)$ is contained in

$$\mathcal{S}(P,\ \max\{r_P,\ d_{PQ} + r_Q + \textit{allowance}\}).$$

Proof. Take $Q + \delta_Q$ with $F(Q + \delta_Q) = 0$ and $\|\delta_Q\| \le r_Q$. The alignment $d_q = \Pi(P - Q - N)$ chosen by `wrap` gives an equivalent solution of the first center, with displacement of norm at most $d_{PQ} + r_Q$ by the triangle inequality. The phase-ball solution set $\mathcal{S}$ in Definition 84 is closed under this equivalence, so it contains the aligned representative. Inflate to $r_P \vee d_{PQ} + r_Q$ and add the outward allowance. □

The implementation uses `mapH` for Φ. Its two index arrays are genuine permutations. In `variants`, the leading two row indices are distinct and `complement` supplies the other four. The six column labels remain attached to six distinct integers when their phase keys are sorted and reordered; the later tail sort and rotation only permute four row labels. With the transpose flag the lookup is $6\,\texttt{col}[j] + \texttt{row}[i]$; either lookup is a bijection of all 36 entries. The sign flag multiplies every entry, and every column of T, by the same sign.

For a finite certified center, `canonicalMap` obtains at least one callback. In each finite pivot list a largest gap is attained and admitted by the nonnegative tolerance. A list attaining the smallest such gap is admitted by the next comparison; among its five tail scores a maximum is admitted by the final comparison. The first resulting callback initializes the chosen map. Ties and repeated values do not alter the distinct attached labels. The signature used to choose among maps need not be a canonical invariant.

`Atlas::find` computes an explicit wrapped dephased difference, projects it, and evaluates its norm. On success it adds 10^{-7} to the computed norm, exceeding the dephasing and projection errors certified by the arithmetic bounds above. `Atlas::add` grows the selected radius to at least the incoming radius plus this distance, with another 10^{-8} added and a separate $2 \cdot 10^{-8}$-margin comparison to the cap. It remains valid when an alternative map found by `variants` supplied the distance; only existence of that allowed map is needed.

If no merge is certified, the source inserts the transformed Root and its original local radius. The hash buckets, pivot selection, and early raw-distance cuts only reduce opportunities to merge; none removes an input Record. The A atlas uses radius cap .125; its size cap aborts rather than certifying a discarded Record.

Proposition 39 (Output of a successful A stage)**.** *Suppose the initial seed cover is valid and the completion search processes all its seeds with no failed or unresolved seed. Then the A-stage Records cover, up to extendibility-preserving equivalence, every exact second-basis Hadamard matrix required for the four-basis problem.*

Proof. The completion parametrization covers every feasible matrix in a seed by its closed unitary cells. Follow any such matrix through containing cells. A sound exclusion cannot reject it; subdivision retains a containing child. Since the search is finite and the failed-setup and unresolved paths are assumed absent, this path must end with a certified local covering bound. The preceding propositions give a certified Root ball for that assignment. Atlas insertions preserve these balls, and every merge preserves their solution sets up to equivalence (Lemma 38). Boundaries cause no exception: overlapping assignments are harmless. □

Finite bounds. The control flow has explicit finite bounds. Before filtering, the nondecreasing five-bin lists for the seed cover number at most $\binom{116}{5} = 160{,}389{,}488$, and for any fixed shared bin the four ordered tail bins number at most $\binom{115}{4} = 6{,}913{,}340$. The former bound is below the signed 32-bit limit. Pair traversals use range loops and 64-bit counters. All bin differences lie between -111 and 111; their nine-coordinate square sum is at most $9 \cdot 111^2$, so the integer distance and bucket arithmetic fit the stated types. Stored five-coordinate differences fit the signed eight-bit type used by the cover.

A retains nonfinal unitary grid lists only through grid 32; grid 64 has at most $64^4/8 = 2{,}097{,}152$ cell tests per seed. Further tests are bounded by the 1,000,000 extra-node cap, and the deepest grid is 512. Reaching the cap or the deepest grid without a certificate records an unresolved seed. Trial-direction and Root proposal failures can lose an opportunity to cover a cell, but cannot make an uncovered cell successful. The local candidate list uses `topK=4` in an array of size 8.

Refining the phase-ball cover.

Stage B replaces a finite collection of certified phase balls by balls of a smaller prescribed radius. Centers need not be exact Hadamard matrices, and no regularity assumption on the Hadamard locus is made. The tangent coordinates used below organize an exhaustive search; the normal estimates from control the remaining directions. A phase matrix is identified with a vector in $\mathbb{R}^{36}$; stars mean real transpose; $\mathcal{V} = \operatorname{im}\Pi$ is the horizontal subspace.

For a certified center P and radius $R \geq 0$ put

$$\mathcal{S}(P,R) = \{[e(P+\delta)] : \delta \in \mathcal{V},\ \|\delta\| \leq R,\ F(P+\delta) = 0\}, \tag{84}$$

where brackets denote the equivalences of Lemma 5 (signed permutations and transpose); these preserve extendibility by two further bases. A cover by such sets is therefore sufficient even when different parts use different arrangements of the same matrix.

Certified relations between centers.

For a Root at P denote its stored tangent matrix $T : \mathbb{R}^k \to \mathbb{R}^{36}$ and put

$$o_P = \mathtt{ort}_P, \quad \alpha_P = \mathtt{id}_P, \quad \gamma_P = \mathtt{C}_P, \quad \mu_P = \mathtt{mnorm}_P, \quad \varepsilon_P = \mathtt{froot}_P, \quad n_P(s) = \mathtt{n0}_P + \alpha_P s + \gamma_P s^2. \tag{85}$$

All these constants are nonnegative. The same notation with subscript Q applies to a second certified Root with center Q and tangent $S : \mathbb{R}^{k_Q} \to \mathbb{R}^{36}$. In particular

$$\|T^*T - I\| \leq o_P, \qquad \|S^*S - I\| \leq o_Q, \qquad \|F(Q)\| \leq \varepsilon_Q. \tag{86}$$

Lemma 40. *Put $A = I - TT^*$. For every $x \in \mathbb{R}^{36}$, with $\zeta = T^*x$,*

$$\|T\| \le \sqrt{1+o_P}, \qquad \|A\| \le 1 + o_P, \qquad \|x\|^2 \le (1+o_P)\|\zeta\|^2 + \|Ax\|^2. \tag{87}$$

These statements do not require x or the columns of T to be horizontal.

Proof. The Gram bound gives the first inequality. The eigenvalues of TT^* lie in $[0, 1+o_P]$, so those of $I - TT^*$ lie in $[-o_P, 1]$; its norm is at most $\max(1, o_P) \le 1 + o_P$. Finally, $\|Ax\|^2 = \|x\|^2 - 2\|\zeta\|^2 + \|T\zeta\|^2 \ge \|x\|^2 - (1+o_P)\|\zeta\|^2$, using $\|T\zeta\|^2 \ge (1-o_P)\|\zeta\|^2$. □

Let D_0 be the unwrapped dephasing operator $(D_0X)_{ij} = X_{ij} - X_{i0} - X_{0j} + X_{00}$. Choose any integer matrix N and define

$$d_q = \Pi(D_0Q - D_0P - N), \qquad \eta_q = T^*d_q, \qquad n_q = \|Ad_q\|. \tag{88}$$

Because $D_0X - X \in \ker\Pi$, $Q - P - d_q \in N + \ker\Pi$; vectors in $\ker\Pi$ are row and column rephasings. Consequently $P + d_q$ and Q differ only by integer phases and rephasings.

Fix a parent radius R and an exact solution displacement $\delta \in \mathcal{V}$ with $\|\delta\| \le R$. Set

$$\eta = T^*\delta, \qquad h' = \delta - d_q, \qquad d = \|h'\|, \qquad \xi = T^*h' = \eta - \eta_q, \qquad \nu = Ah'. \tag{89}$$

Both d_q and h' are horizontal, and $Q + h'$ is equivalent by the same rephasing to $P + \delta$; hence $Q + h'$ is exact. The Root estimates at both centers thus apply to this one solution.

Proposition 41 (Relation estimates)**.** *The following three basic bounds hold.*

$$d \le \min\{R + \|d_q\|,\ \sqrt{(1+o_P)R^2 + n_q^2}\}, \tag{90}$$

$$\|\nu\| \le \min\{\nu_0, (1+o_P)d, \lambda d + e_0\}, \tag{91}$$

$$\|S^*h'\| \le \beta t + \sigma\|\nu\|. \tag{92}$$

Here

$$\nu_0 = n_P(R) + n_q, \qquad \lambda = \alpha_P + \gamma_P(R + \|d_q\|), \qquad e_0 = \mu_P\varepsilon_Q, \tag{93}$$

$$\beta = \sqrt{(1+o_P)(1+o_Q)}, \qquad \sigma = \|S_h\|\,(1+o_P)^{1/2}, \qquad t = \|\eta - \eta_q\|, \tag{94}$$

and $S_h = \Pi S$; and for the next step the child Root's normal and tangent bounds apply to h':

$$d^2 \le n_Q(d)^2 + (1+o_Q)\|S^*h'\|^2, \qquad n_Q(d) = \mathtt{n0}_Q + \alpha_Q d + \gamma_Q d^2. \tag{95}$$

Proof. The triangle inequality gives $d \le R + \|d_q\|$. The identity $\|x\|^2 \le (1+o_P)\|T^*x\|^2 + \|(I - TT^*)x\|^2$ from Lemma 40, applied to h', gives the second inequality in (90).

For (91): $\nu = A(\delta - d_q)$, so $\|\nu\| \le \|A\delta\| + \|Ad_q\| \le n_P(R) + n_q = \nu_0$ by the parent Root normal bound and the definition of n_q. The operator bound $\|A\| \le 1 + o_P$ gives $(1+o_P)d$. For $\lambda d + e_0$, apply (55) of the parent Root to h' with horizontal comparison point d_q: $\|Ah'\| \le [\alpha_P + \gamma_P(\|\delta\| + \|d_q\|)]d + \mu_P\|F(P + d_q)\|$. By rephasing invariance $\|F(P + d_q)\| = \|F(Q)\| \le \varepsilon_Q$, giving $\|Ah'\| \le \lambda d + e_0$.

For (92), first use horizontality of h' to write $S^*h' = S_h^*h'$. The identities $h' = T\xi + \nu$ and $T^*\nu = (I - T^*T)\xi$ yield

$$S^*h' = S_h^*T\xi + S_h^*A\nu + S_h^*T(I - T^*T)\xi.$$

Put $\beta := \sqrt{(1+o_Q)(1+o_P)} \ge \|S_h^*T\|$ and $\sigma := \|S_h\|\sqrt{1+o_P} \ge \|S_h\|\,\|A\|$, using $\|A\| \le 1 \le \sqrt{1+o_P}$. Then $|S_h^*T\xi| \le \beta\|\xi\| = \beta t$, and since $T^*\nu = (I - T^*T)\xi$ gives $\|(I - T^*T)\xi\| \le o_P\|\xi\| \le o_P t$,

the third term $|S_h^* T(I - T^*T)\xi|$ is at most $\beta o_P t$. Absorbing that extra $(1 + o_P)$ factor into β (as the source does), the first and third terms together are at most βt, and the middle term is at most $\sigma\|\nu\|$. This makes explicit why no extra `hor` correction is needed at this step: the horizontal projection appears in S_h once, and both T and $S_h^* T$ are controlled by the Gram bounds.

For (95), apply the squared inequality at the child Root to the horizontal exact-solution displacement h' of Q: $d^2 = \|h'\|^2 \le n_Q(d)^2 + (1 + o_Q)\|S^* h'\|^2$. □

Uniform iteration without contraction.

The relation estimates yield a useful upper iteration without requiring either contraction or a fixed point.

$$\begin{aligned} z &= \sqrt{1 + o_P}\, t, \\ v(b) &= \min\{v_0,\ (1 + o_P)b,\ \lambda b + e_0\}, \\ \psi_P(b) &= \sqrt{z^2 + v(b)^2}, \\ \psi_Q(b) &= \sqrt{(1 + o_Q)\big(\beta t + \sigma v(b)\big)^2 + n_Q(b)^2}. \end{aligned} \tag{96}$$

Start at

$$b_0 = \min\{D,\ \sqrt{z^2 + v_0^2}\}, \tag{97}$$

and replace b_j by

$$b_{j+1} = \min\{b_j,\ \psi_P(b_j),\ \psi_Q(b_j)\}. \tag{98}$$

Lemma 42. *Every iterate in (97)–(98) is an upper bound for the distance d of every parent solution with $\|\eta - \eta_q\| \le t$. The same conclusion holds if any proposed bound is replaced by a larger certified number, or if the process is stopped after any finite number of steps.*

Proof. Both candidates in (97) are upper bounds by Proposition 41. If $d \le b_j$, (91) and nonnegativity of all coefficients give $\|\nu\| \le v(b_j)$. Thus (90) gives $d \le \psi_P(b_j)$; and (92)–(95), together with $n_Q(d) \le n_Q(b_j)$, give $d \le \psi_Q(b_j)$. Their minimum is therefore still an upper bound. The argument holds simultaneously for every solution satisfying the tangent constraint; it does not select a single solution at any step. Enlargement and early stopping preserve the conclusion. □

The rho search for usable tangent thresholds.

Fix a certified parent Root P, T and a reference center Q, S (an atlas child or an earlier blocker) with certified constants $o_Q, \alpha_Q, \gamma_Q, \mu_Q, \varepsilon_Q$. For every tangent threshold $t \ge 0$, let $\widehat{D}(t)$ denote a certified upper bound for $\|h'\| = \|\delta - d_q\|$ over every parent solution δ with $\|\eta - \eta_q\| \le t$. Such a bound is obtained by evaluating (98) at $b_0(t) = \min\{D,\ \sqrt{(1 + o_P)t^2 + v_0^2}\}$; by Lemma 42, the final iterate is a valid upper. The source stores it back through the alignment (88).

$$\|\eta - \eta_q\| \le t \text{ and } \widehat{D}(t) \le a \implies [e(P + \delta)] \in \mathcal{S}(Q, a) \quad \text{for every parent solution } \delta. \tag{99}$$

For a queried sphere radius $0 \le a \le .126$, the source's `init` searches for a usable tangent distance ρ as follows. Set $\ell = 0$, $u = 1.01a$. If $\widehat{D}(0) > a - 10^{-7}$, report no relation. Otherwise make 25 midpoint trials, each with $m = (\ell + u)/2$: if $\widehat{D}(m) < a - 2 \cdot 10^{-7}$, replace ℓ by m; otherwise replace u by m. Finally set

$$\rho = \max(0, \ell - 10^{-8}), \tag{100}$$

and report a usable relation only if $\rho > 10^{-7}$.

Lemma 43. *Every usable returned threshold certifies*

$$\|\eta - \eta_q\| \le \rho \quad \Longrightarrow \quad [e(P+\delta)] \in \mathcal{S}(Q, a)$$

for every parent solution δ. The search need not find the largest possible threshold; in particular its correctness does not require monotonicity of the numerical function $\widehat{D}$.

Proof. A positive returned $\rho > 10^{-7}$ cannot arise from the initial value $\ell = 0$. Thus the final stored ℓ was assigned by a successful trial $\widehat{D}(\ell) < a - 2 \cdot 10^{-7}$, and $\rho = \ell - 10^{-8} < \ell$. Take any parent solution δ with $\|\eta - \eta_q\| \le \rho \le \ell$. By construction $\widehat{D}(\ell)$ is an upper bound for $\|h'\|$ at this solution, so $\|h'\| < a - 2 \cdot 10^{-7} < a$. Apply (99). No comparison between $\widehat{D}$ at distinct arguments is needed. □

For a previously completed parent used as a blocker, a is its fixed input radius and this lemma suffices. For an atlas child, the search uses the allowed cap $a = \tau$; when boxes are assigned to that child, the code evaluates $\widehat{D}$ again at an upper tangent distance for those boxes and grows the stored radius to that certified distance, with an additional margin. It then checks both that the tangent argument is at most ρ and that the new radius is at most τ; failing either check aborts. Thus even an unfavorable rounded reevaluation cannot silently assign a box to a sphere too small to contain it.

Exhaustive grid in tangent coordinates.

For every parent solution in the parent record,

$$\|\eta\| = \|T^*\delta\| \le \sqrt{1 + o_P}\, R =: E_P. \tag{101}$$

Let $w > 0$ be the stored refinement width regarded as an exact real number. The code forms an inflated upper bound E for E_P by adding 10^{-7} to the computed right side of (101), and chooses

$$L = \lceil E/w + 0.500001 \rceil, \qquad n = 2L + 1,$$

with the stated rounded evaluation. The arithmetic of ensures that the outer endpoints $\pm(L+1/2)w$ contain $[-E_P, E_P]$.

For $u \in \mathbb{R}^k$ and $h \ge 0$ write

$$B(u, h) = \prod_{a=1}^{k} [u_a - h, u_a + h]. \tag{102}$$

The base boxes have $u_a = b_a w$, $-L \le b_a \le L$, and $h = w/2$. Their union contains $[-E_P, E_P]^k$, so every tangent vector in (101) belongs to at least one base box. All intervals are closed; grid boundaries are included.

Blocker induction.

A *blocker* is an earlier input parent whose entire refinement has succeeded. It is queried at its original fixed radius. An atlas child, in contrast, is queried directly as a certified-center sphere and its radius is grown when needed. A child created during an unfinished parent search can already be reused; the unfinished parent itself cannot become a blocker.

Proposition 44. *Suppose Stage B runs on a finite list of records with valid Root certificates, under the arithmetic assumptions and error bounds described above. On a completed run, let $\mathcal{A}$ be the final atlas of children and $\mathcal{B}$ the list of input parents whose refinement failed. Then*

$$\bigcup_{(P,R)\ \text{input}} \mathcal{S}(P, R) \subseteq \bigcup_{(Q,r)\in\mathcal{A}} \mathcal{S}(Q, r) \cup \bigcup_{(P,R)\in\mathcal{B}} \mathcal{S}(P, R). \tag{103}$$

In particular, a completed run with no bad parents replaces the input cover by a child cover. A nonempty bad list remains an obligation.

Proof. Process the parents in the order used by the program. The induction invariant is that every previously successful parent sphere is covered by the current atlas; it holds trivially before the first parent, and appending children or increasing radii preserves it.

For the current parent, first consider an individual box in its finite subdivision tree. A successful outside test proves that the box contains no parent solution. A successful direct assignment covers all its solutions by Lemmas 32 and 33. A successful relation to a child covers them by that child's checked radius update. A successful relation to a blocker covers them by the blocker sphere, which is already covered by the atlas by the induction hypothesis. Finally success on all children covers the box by inclusion of their union. These are all the successful returns of the cell solver, so induction from the leaves proves its assertion for every successful cell.

The grid covers every possible tangent vector by (101). The traversal can finish successfully only after every base box has either been rejected as outside, marked by a valid whole-box relation, or solved as above. Hence success for the current parent proves coverage of its entire sphere, permitting its addition to the blocker list. If the parent fails, it is not added and is retained in $\mathcal{B}$. Any children produced while attempting it remain valid stored spheres and are reused without assuming the failed parent has been covered. □

This proof involves two finite inductions — on subdivision depth and on the order of completed parents — and neither infers coverage of a parent from children that merely assume coverage of that same parent. Later radius growth cannot invalidate an earlier certificate, because centers and Root constants are immutable and the corresponding solution sets only enlarge.

Batch files and finite failure conditions.

Batching writes appended atlas Roots in their insertion order, in groups of at most 256. The radius written for each Root is the common cap τ rather than its current achieved radius. If a Root at index i is emitted before the run ends, every later state satisfies

$$r_i \le \tau, \qquad \mathcal{S}(Q_i, r_i) \subseteq \mathcal{S}(Q_i, \tau), \tag{104}$$

with exactly the same center, tangent columns, and certificate constants as in its batch. No future merge changes these data. At successful completion the final partial batch and the count marker account for every appended index. Replacing the child union in (103) by the union of all batch spheres at radius τ is therefore valid.

For clarity, the implementation uses the following finite limits. Input radii satisfy $0 \le R \le 0.126$, and the child cap and width satisfy

$$0.0001 \le \tau \le 0.126, \qquad 0.0001 \le w \le .05, \qquad 0 \le k \le 10.$$

Normal constants are checked against

$$\gamma_P < 20, \quad \mathtt{n0}_P, \varepsilon_P < 10^{-6}, \quad \alpha_P < 10^{-4}, \quad o_P, \mathtt{hor}_P < 10^{-5}, \quad \mu_P < 2.$$

Stored phase entries have magnitude at most 4, and stored tangent entries at most 1.01. These range checks accompany genuine Root certificates; the checks alone do not prove those certificates. A prospective Root with more than ten tangent columns is rejected, not truncated.

The base grid is limited to 50,000,000 boxes, subdivision depth at most six, atlas size at most 2,000,000 Roots. One record file holds at most 2,000,000 records and the combined input at most

8,000,000. The grid product is checked before multiplication past its limit. Within the admitted ranges $L \le 1262$; the mixed-radix indices fit the stated integer types, and every split uses at most 2^{10} children. The integer and floating-point bounds are recorded in the arithmetic bounds above.

Excluding two further mutually unbiased bases.

The final stage takes a certified Root (P, T) with $0 \le r \le .03$ and rules out two more mutually unbiased orthonormal bases in the phase ball of radius r. The phase variables of the two extra bases are divided into closed balls, a permissive graph of necessary edge relations is retained, and a finite search runs for the required clique pattern. Every rejection is a strict violation of a necessary condition with a certified margin; no rejection depends on a center being exact.

Throughout, $e(t) = \exp(2\pi i t)$. A real matrix in $\mathbb{R}^{36}$ is identified with row-major indexing. The Root constants $\omega, n_0, \iota, \Gamma$ denote `ort`, `n0`, `id`, `C` as in the arithmetic bounds above. Its certified data give

$$\|T^*T - I\| \le \omega, \qquad \|F(P)\| \le f_0 < 10^{-6}, \qquad 0 \le \omega < 10^{-5}, \tag{105}$$

and, for every horizontal δ with $\|\delta\| \le r$ such that $e(P + \delta)$ is Hadamard,

$$\eta = T^*\delta, \quad \nu = \delta - T\eta, \qquad \|\eta\| \le r_H := \sqrt{1+\omega}\, r, \qquad \|\nu\| \le n := n_0 + \iota r + \Gamma r^2. \tag{106}$$

The first two bases are the coordinate basis and the columns of $e(P + \delta)/\sqrt{6}$. If further bases exist, they share this single Hadamard displacement δ, hence this single η. Each vector unbiased to the coordinate basis is $e(\beta)/\sqrt{6}$ for some real phase vector β; choose its global phase so that $\beta_0 = 0$, and lift $\beta_1, \dots, \beta_5$ to $[0, 1]$. Different $U(1)$ rephasings of different extra vectors do not change δ.

The single-vector sums use a uniform operator bound. The identity $\|AA^* - 6I\|_F = 2\pi\sqrt{2}\,\|F(P)\|$ and (105) give

$$\|A/\sqrt{6}\|_{\mathrm{op}} \le \sqrt{1 + \frac{2\pi\sqrt{2}}{6} f_0} < K, \qquad K := 1.00001, \tag{107}$$

where $A = e(P)$.

Closed phase bins.

At resolution M the five variable phases are covered by closed intervals $[b/M, (b+1)/M]$, $0 \le b < M$. Phase 1 equals 0 on the torus, so this covers every boundary. Let u be a bin midpoint, with $u_0 = 0$, and let

$$a = \frac{1}{2M}, \quad d_0 = 0, \ |d_j| \le a \ (j > 0), \quad x = d - \frac{1}{6}\Big(\sum_j d_j\Big)\mathbf{1}.$$

Then the bin vector equals $e(u + x)/\sqrt{6}$ up to global phase.

Let $S_i = \sum_j e(P_{ij} - P_{i0})$, the (unscaled, global-phase-free) ith entry of the row over inner products. Set

$$c_i = \frac{|S_i|^2 - 1}{4\pi}, \qquad J_{ij} = -\operatorname{Im}(\overline{S_i} q_{ji}), \qquad B_{ik} = \sum_j J_{ij} T_{(j,i),k}, \tag{108}$$

where $q_{ji} = e(P_{ij} - P_{i0})$ (so $\sum_j q_{ji} = S_i$; and note $\sum_j J_{ij} = 0$). For $y \in \mathbb{R}^6$ put

$$G_y := \max_j \Big|\sum_i y_i \overline{S_i} q_{ji}\Big|, \qquad N_y := \Big(\sum_{i,j} y_i^2 J_{ij}^2\Big)^{1/2}.$$

Lemma 45 (Direct displacement tests)**.** *If $\|x\| \le X$, $\|x\|_1 \le L$, and $\|\delta\| \le r$, then*

$$\|S' - S\| \le 2\pi(KX + r), \qquad |S'_i - S_i| \le 2\pi(L/\sqrt{6} + r). \tag{109}$$

Consequently unbiasedness requires

$$\max\{0, 1-\tau\}^2 \le |S_i|^2 \le (1+\tau)^2, \qquad \sum_i (|S_i| - 1)^2 \le [2\pi(KX + r)]^2, \tag{110}$$

with $\tau = 2\pi(L/\sqrt{6} + r)$.

Proof. First perturb the vector phases by x and then the Hadamard phases by δ. The first change has norm at most $K\|(e(x_j) - 1)_j\| \le 2\pi KX$. For the second, the ith component has magnitude at most $(2\pi/\sqrt{6}) \sum_j |\delta_{ji}| \le 2\pi\|\delta_{:,i}\|$, and their combined norm is at most $2\pi r$. The scalar bound is the same with the first change at most $2\pi L/\sqrt{6}$. Apply $||S_i| - |S'_i|| \le |S_i - S'_i|$ and use $|S'_i| = 1$. □

Lemma 46 (Weighted Taylor estimate)**.** *Suppose $\|x\| \le X$ and $\|x\|_\infty \le I$. For every $y \in \mathbb{R}^6$, unbiasedness implies*

$$|y \cdot (c + Jx - B\eta)| \le \mathcal{E}(y; X, I), \tag{111}$$

where

$$\begin{aligned}\mathcal{E}(y; X, I) := nN_y + \pi\Big[&\|y\|_\infty (KX + r)^2 + \min\{G_y X^2,\ K\|S \circ y\| IX\} \\ &+ \frac{2}{\sqrt{6}}\|S \circ y\| Xr + \frac{1}{\sqrt{6}}\|S \circ y\|_\infty r^2\Big].\end{aligned} \tag{112}$$

For a bin, this gives the necessary support inequality

$$|y \cdot c| \le a \sum_{j>0} |(J^* y)_j| + r_H \|B^* y\| + \mathcal{E}(y; X_M, I_M). \tag{113}$$

Proof. Write $\Delta_i = S'_i - S_i$. The expansion of $(|S'_i|^2 - 1)/(4\pi)$ has constant term c_i, linear term $\sum_j J_{ij}(x_j - \delta_{ji})$, and the squared increment $|\Delta_i|^2/(4\pi)$. After multiplication by y_i and summation the latter is bounded by $\pi\|y\|_\infty (KX + r)^2$ via (109).

The pure-x linear remainder was bounded in the preceding scalar Taylor expansion; its two alternatives are $G_y X^2$ and $K\|S \circ y\| IX$, and the minimum is valid. The Hadamard-δ linear term is $B\eta$; its remainder with ν is bounded by nN_y. The cross term and pure δ term give the remaining $\|S \circ y\| Xr$ and $\|S \circ y\|_\infty r^2$ contributions. Subtracting the bin's linear box term yields the necessary support bound (113). □

The bin routine `Layer::test` applies (110), then the scalar test (114) (see below), and finally (113) for proposed directions. Proposed directions come from an inexact linear solve; the proposal method is immaterial because the final inequality holds for every real y and is checked at the stored test direction.

There is a useful scalar test before the tangent decomposition. Set $S_{\max} = \max_i |S_i|$, and for a bin write $X = X_M$, $L = L_M$, $I = I_M$. Unbiasedness requires

$$|c_i| \le a \sum_{j>0} |J_{ij}| + r\|J_{i,:}\| + \rho, \tag{114}$$

where

$$\rho = \min\Big\{\pi\big[(KX+r)^2 + S_{\max}(KIX + (2Xr+r^2)/\sqrt{6})\big],$$
$$\pi\big[(L/\sqrt{6}+r)^2 + S_{\max}(X+r)^2/\sqrt{6}\big]\Big\}. \tag{115}$$

The first remainder bound follows from the weighted estimate with a coordinate direction and without decomposing δ. The second uses the scalar increment bound in (109) and applies the second-order exponential bound directly to $x-\delta_{:,i}$, whose norm is at most $X+r$. The full Hadamard linear term is supported by $r\|J_{i,:}\|$, so no normal-error term is added.

The implementation checks that a proposed direction is finite with maximum absolute entry strictly between 10^{-100} and 10^{100}, normalizes by that maximum, and uses the resulting stored coefficients as the test direction. The stored direction itself is what matters.

From bins to closed vector balls.

The initial resolution is $M = 12$. Every bin is tested, and each retained bin is split into all $2^5 = 32$ children at each doubling $12 \to 24 \to 48 \to 96$. On one coordinate, the child indices $2b$ and $2b+1$ cover the parent interval exactly, so every vector follows at least one surviving bin path, including on a phase or torus boundary. Exceeding the cap of 2,000,000 retained bins is an unresolved outcome, never an exclusion. In the stated invocation there are four bin layers.

For $v \in \mathbb{R}^6$ and $s \geq 0$, define the closed vector ball

$$\mathcal{B}_{\mathrm{vec}}(v,s) = \Big\{\frac{\lambda e(v+x)}{\sqrt{6}} : |\lambda| = 1,\ x \in \mathbb{R}^6,\ \sum_j x_j = 0,\ \|x\| \leq s\Big\}. \tag{116}$$

Lemma 47 (Certified clustering)**.** *Suppose each retained bin midpoint u is assigned to a center v and an integer vector $m \in \mathbb{Z}^6$. Let*

$$h = u - v - m - \frac{1}{6}\Big(\sum_j (u_j - v_j - m_j)\Big)\mathbf{1}.$$

If the assigned radius is at least $\|h\| + X_M$, then $\mathcal{B}_{\mathrm{vec}}(v,s)$ covers the whole bin up to global phase. The final assignment in `cluster` *has this property.*

Proof. For any raw displacement d in the bin, subtracting its mean gives the x of the direct displacement representation. With the assigned integers fixed, the phase difference from v is $h+x$ up to an integer vector and a scalar phase. It has zero mean and norm at most $\|h\| + X_M$, and every bin boundary is attained by some representative of the closed bin, so the covering assertion also holds on every boundary. □

Each assignment iteration computes coordinatewise integer wraps of $u - v$, then subtracts their mean. A candidate rejected by its distance cutoff is simply not used; the bin is assigned to another center or to a newly inserted midpoint. Thus no bin is dropped by the heuristic. For each assigned center the radius is updated to the maximum of the computed projected midpoint distance plus the bin-norm upper bound and 10^{-9}; this is an outward upper bound as certified in the arithmetic bounds above. New midpoint centers use midpoint distance zero, with the same allowance.

The earlier Lloyd iterations adapt the nearest-center/centroid heuristic of [17, Sections IV–VI]. They may move centers, but a subsequent complete assignment follows each such movement. In the last iteration the assigned centers are not moved. Deleting centers with no assigned bins cannot lose coverage. The returned centers and radii therefore meet the covering assertion. An attempt to

exceed 2048 centers, or a failure of the final checks $|v_j| \le 2$ and $0 \le s < .1$, is unresolved rather than negative.

For the specified invocation, the heuristic radius parameter is .042 and there are two assignment iterations. With $w = 1/M$ the candidate threshold is $(.042 - 1.11w)^2$, and the coordinate cutoff is the square root of twice that threshold plus 10^{-10}. These constants affect only the choice of assignments; the proof is the final radius update in Lemma 47. In particular, the argument does not assume that the selected wrapped lift minimizes torus distance.

Lemma 48 (Distinct labels)**.** *Two vectors in the same ball $\mathcal{B}_{\mathrm{vec}}(v, s)$ with $s < .1$ are neither orthogonal nor mutually unbiased. Hence the twelve vectors in two further mutually unbiased orthonormal bases must occupy twelve distinct retained balls.*

Proof. After removing global phases, their unscaled inner-product sum has the form $\sum_j e(z_j)$ with $\|z\| \le 2s < .2$. Therefore

$$\Big|\sum_j e(z_j)\Big| \ge 6 - \sum_j |e(z_j) - 1| \ge 6 - 2\pi\sqrt{6}\,\|z\| > 6 - \frac{2}{5}\pi\sqrt{6} > \frac{20}{7} > \sqrt{6}.$$

Orthogonality requires the unscaled magnitude to be zero, and unbiasedness requires it to be $\sqrt{6}$. Every pair among the twelve has one of these two relations, so no pair shares its ball. □

Bin geometry and edge tests.

Lemma 49 (Bin geometry)**.** *Fix a bin with midpoint $u \in \mathbb{R}^6$ ($u_0 = 0$) and halfwidth $a = 1/(2M)$. Any phase vector in the bin is $u + d$ with $d_0 = 0$ and $|d_j| \le a$ for $j \ge 1$. The horizontal displacement $x = d - \frac{1}{6}(\sum_j d_j)\mathbf{1}$ satisfies $\sum_j x_j = 0$ and, with the constants X_M, L_M, I_M of ,*

$$\|x\| \le X_M, \quad \|x\|_1 \le L_M, \quad \|x\|_\infty \le I_M. \tag{117}$$

Proof. $\sum_j x_j = 0$ is immediate from the definition. Write $d = \bar{d} + (d - \bar{d})$ with $\bar{d} = (1/6)\sum_j d_j$. Then $\|x\|^2 = \|d - \bar{d}\mathbf{1}\|^2 = \|d\|^2 - 6\bar{d}^2 \le \|d\|^2 \le 5a^2$, so $X_M \ge \sqrt{5}\,a$ suffices. Each $|d_j - \bar{d}| \le 2a$, so $L_M \ge 10a$ and $I_M \ge 2a$ also suffice. accounts for the 10^{-12} margin. □

Take two distinct balls with centers u, v and radii $s_u, s_v < .1$. For actual representatives with displacements x_u, x_v, set

$$Q_j = e(u_j - v_j), \quad S = \sum_j Q_j, \quad z = x_u - x_v, \quad r_e = s_u + s_v, \quad \|z\| \le r_e. \tag{118}$$

The actual unscaled inner product is, up to conjugation and global phase, $S' = \sum_j Q_j e(z_j)$. Since $|S' - S| \le 2\pi\sqrt{6}\,r_e$, necessary direct tests for orthogonality ($\mathcal{O}$) and unbiasedness ($\mathcal{U}$) are

$$\frac{|S|}{2\pi\sqrt{6}} \le r_e \ (\mathcal{O}), \qquad \frac{\big||S|/\sqrt{6} - 1\big|}{2\pi} \le r_e \ (\mathcal{U}). \tag{119}$$

$$(D_{\mathcal{O}})_{:,j} = \begin{pmatrix} -\operatorname{Im} Q_j \\ \operatorname{Re} Q_j \end{pmatrix}, \qquad (D_{\mathcal{U}})_j = \frac{\operatorname{Re} S\,(D_{\mathcal{O}})_{1j} + \operatorname{Im} S\,(D_{\mathcal{O}})_{2j}}{6}. \tag{120}$$

Lemma 50 (Edge Taylor estimates)**.** *For any actual pair from two distinct closed vector balls, with the data of* (118)*, and for* $S_{\mathcal{O}} = \operatorname{Re} S$*,* $S_{\mathcal{U}} = (|S|^2 - 6)/(4\pi)$*,*

$$|\operatorname{Re}(S' - S) - (D_{\mathcal{O}} z)_1| \le E_{\mathcal{O}} := 4\pi\sqrt{6}\, r_e^2, \tag{121}$$

$$\Big|\frac{|S'|^2 - 6}{4\pi} - S_{\mathcal{U}} - D_{\mathcal{U}} z\Big| \le E_{\mathcal{U}} := \frac{2\pi \max(|S|, \sqrt{6})}{\sqrt{6}}\, r_e^2 + 4\pi r_e^2. \tag{122}$$

Therefore the necessary support conditions

$$|S_{\mathcal{O}}| \le r_e \|D_{\mathcal{O}}\|_F + E_{\mathcal{O}}, \tag{123}$$

$$|S_{\mathcal{U}}| \le r_e \|D_{\mathcal{U}}\| + E_{\mathcal{U}} \tag{124}$$

hold for every actual pair.

Proof. Write $e(z_j) - 1 = 2\pi i z_j + \varepsilon_j$ with $|\varepsilon_j| \le 2\pi^2 z_j^2$ by the integral Taylor remainder. Multiply by $Q_j = a_j + ib_j$ and separate real and imaginary parts: $\operatorname{Re}(Q_j(e(z_j) - 1)) \approx -2\pi b_j z_j = (D_{\mathcal{O}} z)_j$ and $\operatorname{Im}(Q_j(e(z_j) - 1)) \approx 2\pi a_j z_j$ up to a residual of modulus at most $4\pi^2 r_e^2$ per coordinate. Summing over j and using $\|z\| \le r_e$ proves (121).

For unbiasedness, expand $(|S'|^2 - 6)/(4\pi)$ at S. The constant is $S_{\mathcal{U}}$, the linear term is $D_{\mathcal{U}} z$ (the division by 4π normalizes the derivative exactly), and the quadratic remainder is at most $(2\pi|S|/\sqrt{6})r_e^2 + 4\pi^2 r_e^2$; substituting the residual of the orthogonality calculation gives (122). Cauchy–Schwarz on $\|z\| \le r_e$ gives the support inequalities. □

The joint test with one shared tangent variable.

For each ball $t = 0, 1$, form c^t, J^t, B^t, S^t from (108) at its actual center. Write its radius as s_t and put

$$\mathcal{E}_t^{\mathrm{ball}}(y) := n\Big(\sum_{i,j} y_i^2 (J_{ij}^t)^2\Big)^{1/2} + \pi\Big[\|y\|_\infty (K s_t + r)^2 + s_t^2 G_y^t + \frac{2}{\sqrt{6}}\|S^t \circ y\| s_t r + \frac{1}{\sqrt{6}}\|S^t \circ y\|_\infty r^2\Big]. \tag{125}$$

This is exactly the weighted estimate (112) restricted to its first pure-x remainder (do not minimize against the second one here), with $X = s_t$ and $I = s_t$; they are valid on the whole closed Euclidean vector ball because the bound was derived for the worst case over that ball.

For an edge type $E \in \{\mathcal{O}, \mathcal{U}\}$ write $\ell = 2$ or 1 for the dimension of D_E, and let $q_E \in \mathbb{R}^\ell$ be the constant part of the linearized edge equation:

$$q_{\mathcal{O}} := (\operatorname{Re} S, \operatorname{Im} S) \in \mathbb{R}^2, \qquad q_{\mathcal{U}} := \frac{|S|^2 - 6}{4\pi} \in \mathbb{R}^1, \tag{126}$$

with D_E and E_E as in (120) and (121)–(122).

Lemma 51 (Joint necessary support)**.** *Suppose one vector from each of the two balls is unbiased to the coordinate basis and to the* same *second basis, whose phase matrix is* $e(P + \delta)/\sqrt{6}$*, and the two vectors have relation* $E \in \{\mathcal{O}, \mathcal{U}\}$*. For every* $y_0, y_1 \in \mathbb{R}^6$ *and* $y_e \in \mathbb{R}^\ell$*,*

$$\begin{aligned} &|y_0 \cdot c^0 + y_1 \cdot c^1 + y_e \cdot q_E| \\ &\quad \le \sum_{t=0}^{1}\Big[s_t\|(J^t)^* y_t + (-1)^t D_E^* y_e\| + \mathcal{E}_t^{\mathrm{ball}}(y_t)\Big] \\ &\qquad + r_H\|(B^0)^* y_0 + (B^1)^* y_1\| + E_E\|y_e\|. \end{aligned} \tag{127}$$

Proof. For each single vector the linearized equation is $c^t + J^t x_t - B^t \eta$ with weighted residual bounded by $\mathcal{E}_t^{\mathrm{ball}}(y_t)$. The edge equation is $q_E + D_E(x_0 - x_1)$ with residual norm at most E_E.

Divide the edge equation by y_e, expand each x_t-dependent part for the two balls separately, and use the x-support of each single vector's displacement: for ball $t = 0$ the combined coefficient of x_0 is $(J^0)^* y_0 + D_E^* y_e$; for $t = 1$, $(J^1)^* y_1 - D_E^* y_e$. These support over $\|x_t\| \le s_t$ give the first two norm terms.

The single-vector coefficients of the shared tangent η are $-B^0$ and $-B^1$, and combine to give the coefficient $-(B^0)^* y_0 - (B^1)^* y_1$. Support this over $\|\eta\| \le r_H$. Bound the three residuals by the triangle inequality and Cauchy's inequality. □

This is exactly the comparison implemented by `graph::joint`. It first calls `single::support` with a zero vector-displacement coefficient array for each ball, and only then adds the edge derivatives. Thus the maximum coefficient in (125) is formed from the pure single-vector remainder; it cannot be decreased by cancellation with an edge coefficient. The tangent coefficient array, in contrast, is shared and accumulates both $-(B^t)^* y_t$ contributions before its norm is taken.

As with the single-vector test, the proposed direction comes from an inexact linear solve; only its final inequality must be certified. The final inequality holds for every real y; the numeric choice of direction is not part of the certificate.

The inequalities just proved are necessary in exact real arithmetic. In the implementation, each rejection is a strict violation of an outward-enlarged bound; equalities and unresolved comparisons retain the candidate. certifies these comparisons, and lists their margins. Thus every genuine pair retains its required relation when the tests are evaluated numerically.

Exact graph exclusion.

Build one vertex for each retained ball. Join two distinct vertices by an $\mathcal{O}$ edge iff their orthogonality type has not been excluded by the direct and joint tests above; similarly for $\mathcal{U}$ edges and unbiasedness. Both graphs are undirected and have no self-loops. Every genuine pair retains its corresponding edge by the preceding lemmas. Extra edges are harmless: retained pairwise possibilities need not have a simultaneous realization.

By Lemma 47 and Lemma 48, two additional mutually unbiased orthonormal bases would give disjoint vertex sets A, B satisfying

$$|A| = |B| = 6, \qquad A, B \text{ are } \mathcal{O}\text{-cliques}, \qquad A \times B \subseteq E(\mathcal{U}). \tag{128}$$

The following finite search tests this necessary configuration.

Lemma 52 (Complete graph search)**.** *The mode-*1 *search of* `graph::exist`*, started with both pools equal to the full vertex set and first-clique target size six, returns false only if* (128) *is impossible. A work-cap exhaustion never gives a negative certificate.*

Proof. First consider `clique` on a candidate set P and target size m. For $m = 0$ it succeeds, and for $m = 1$ it tests nonemptiness. For $m \ge 2$ it repeatedly selects the least remaining vertex i, removes it from P, and searches $P \cap N_{\mathcal{O}}(i)$ for an $(m-1)$-clique. For $m = 2$ that search is just nonemptiness. Every clique has a unique least vertex, so this enumerates every possibility. Terminating when fewer than m candidates remain is also exact.

Now suppose A, B satisfy (128). Interchange their names if necessary so that $a = \min(A \cup B)$ belongs to A. At the root, `exist` tries possible least vertices of the first clique in increasing order. It deletes each tried root vertex from both its local first pool and its second pool. Thus when a is selected, every required remaining vertex in $A \cup B$ is still available. This root deletion imposes only the chosen convention that the minimum of the union lies in the first clique, and it is performed only

when the remaining first-clique target is six. Recursion then runs `clique` on the first-pool target six, and for each resulting candidate A, on the second-pool target six. When `clique` succeeds, its exact enumeration is used to test $A \times A' \subseteq E(\mathcal{U})$ by checking every cross pair in order.

The search cap and any other finite work bound turn into failure, not into a claim that the pattern is impossible. Thus a work-cap exhaustion is not a negative certificate; only a completely enumerated negative result is. □

Proposition 53 (Local exclusion)**.** *Suppose*

1. *the retained bins cover the closed phase ball of the Root in the sense of Lemma 47;*
2. *every retained bin has either been excluded by (110), (114), or (113), or been accepted and covered by the assigned ball;*
3. *the $\mathcal{O}$- and $\mathcal{U}$-graphs on the retained balls, after excluding any pair by the direct or joint tests conditional on η, admit no configuration (128) as decided by a completed `graph::exist` search.*

Then the phase ball contains no two additional mutually unbiased orthonormal bases.

Proof. Suppose two such bases existed. Each of their twelve vectors is covered by some retained ball (Lemma 47). Lemma 48 puts them in twelve distinct balls. Within each basis the six vectors are pairwise orthogonal, and every intra-pair survived its $\mathcal{O}$ direct and joint tests because the tests are necessary conditions; thus they are joined by $\mathcal{O}$ edges. Across the two bases every pair is unbiased, and similarly they survive the $\mathcal{U}$ tests, so the cross-pairs are $\mathcal{U}$ edges. This produces a configuration (128), contradicting the third hypothesis. □

The conclusion of Proposition 53 is conditional on a completed negative result for the individual Record; no collection of unexecuted tests is itself such a result.

Implementation margins and invocation.

For a bin the source forms its halfwidth with a 10^{-12} increase, then increases each of X_M, L_M, I_M by a further 10^{-12} after the indicated multiplication. The stored tangent and normal bounds are

$$\widehat{r}_H = \sqrt{1+\omega}\, r + 10^{-9}, \qquad \widehat{n} = n_0 + \iota r + \Gamma r^2 + 10^{-9}.$$

The upper squared-magnitude threshold in (110) is increased by $2 \cdot 10^{-7}$, and its lower threshold, when $\tau < 1$, is decreased by that amount. The combined squared-distance threshold, the right side of (114), and the right side of (113) each receive a $2 \cdot 10^{-7}$ comparison allowance.

For a pair, the radius sum $s_u + s_v$ is increased by 10^{-10}. The distance tests (119) permit a further $2 \cdot 10^{-7}$ on that radius. Both edge residual bounds $E_{\mathcal{O}}$ and $E_{\mathcal{U}}$ are increased by $2 \cdot 10^{-8}$; the direct comparisons (123) and (124) receive another $2 \cdot 10^{-7}$. The right side of the joint comparison (127) receives $3 \cdot 10^{-7}$. Cluster assignment radii include the 10^{-9} allowance described in Lemma 47.

The bounds $|P_{ij}| \le 4$ and $|v_j| \le 2$ keep all trigonometric arguments used here within the guarded range of `expair`. The proof of includes the errors in these arguments, in q, S, c, J, B, in the short weighted sums and norms, and in the positive support expressions. Inexact solution of the proposal systems supplies no separate assumption.

For clarity, the invocation parameters used for this certificate are `12 4 1 .042 1 1 5000000`: initial resolution 12, four bin layers, one proposed bin direction, clustering parameter .042, one proposed joint direction, graph mode 1, and the stated graph work cap. The two Lloyd iterations are the fixed default. The conclusion of Proposition 53 is conditional on a completed negative result for the individual Record; no collection of unexecuted tests is itself such a result.

Finite execution and the recorded certificate.

The computation used in the conclusion of this proof was completed on 4 October 2026 (UTC). This section specifies its finite inputs, the tests for complete execution, and the implication from those tests to nonexistence. The geometric and arithmetic justification of each rejection lies in the seed cover, the completion and refinement stages, the exclusion tests, and the certified arithmetic. A successful exit by itself is not the certificate: the inputs, record counts, saved failures, and coverage of all shards must agree as described here.

Programs, records, and machine arithmetic.

The recorded platform was x86-64, Intel Xeon Platinum 8573C, Ubuntu GCC 13.3.0, and glibc 2.39. Each stage was compiled separately, all four on the same UTC date as the run, with contraction disabled and a fixed language dialect. The arithmetic assumptions are binary64, round to nearest, separate operations without contraction or fast-math reassociation, correctly rounded binary64 square root, masked floating-point exceptions, and neither flush-to-zero nor denormals-are-zero. Separate probes compiled with the cover and A/B/C flag sets both passed and recorded `FLT_EVAL_METHOD=0`, 32-bit `int`, 64-bit `size_t`, 64-bit `uint64_t` equal to `unsigned long`, eight-bit bytes, and MXCSR `0x1f80`. Both probes recorded nearest rounding on entry and during the check, and passed the square-root and gradual-underflow samples. These probes and the program checks provide limited evidence for the stated platform conditions, not a proof of all primitive-operation or compiler semantics. Those semantics remain assumptions; the certified arithmetic supplies the numerical bounds under them.

Four programs carry out the search: one builds the seed cover, one runs the completion stage, one refines the resulting phase balls, and one performs the exclusion tests. They share the Root, Atlas and graph data structures through common declarations, and the arithmetic bounds below fix every floating-point operation any of them performs. The archived sources, identified by their recorded digests, are what make the certificate concrete rather than notional.

How the pipeline is split.

The first program writes a seed-center table, one entry per retained midpoint, carrying its rational phases in cycles, the selected column position s, and the radius 0.032. It performs only the seed filters whose margins are certified in the arithmetic bounds.

The completion stage processes seeds in disjoint shards and fills the atlas of certified phase balls. The refinement stage then shrinks that cover and writes its batch files, and the exclusion stage tests every resulting ball for two extra bases, writing a per-batch record of the few that resist exclusion.

Each program reports its own exit status and counts. A supervising driver records one outer summary and one summary per stage, reruns the fallback and the final exclusion shards whenever the initial exclusion pass leaves any record unexcluded, and compares every intermediate count against the outer summary before the run counts as complete.

Control flow and terminal conditions.

The control flow is finite and every terminal condition is documented. A run that fails any listed condition is not a certificate.

The required terminal conditions are:

1. successful builds, a normally completed cover, and a complete center file;
2. four A exits 0, all assigned centers processed, terminal `failedBalls=0`, and four empty bad text files;
3. both B exits 0, all input parents processed, agreement of logged and saved output counts, and zero-count B bad files;

4. every initial C batch processed exactly once with a consistent exit, done count, bad count, and saved record count;

5. every initial C bad record is included in fallback, fallback exits 0 with all parents processed and zero saved failures, and the four final C exits are 0 with total done equal to the fallback count and zero saved failures.

Missing files, truncated data, an unexpected exit, resource exhaustion, an unresolved A or B parent, or a final C bad record prevent acceptance. Recoverable search diagnostics are distinguished from these terminal conditions. For example, A's `ownfail` counts unsuccessful proposals of a local Root cover that are followed by subdivision. A0 reported such a diagnostic in this run; its terminal unresolved count was nevertheless zero.

The finite bounds governing this control flow — the bin counts, the range of stored differences, the grid 64 cell-test count, and the extra-node and deepest-grid caps — are stated and checked in the completion stage.

Reaching the C-stage bin cap of 2,000,000 retained bins, exceeding the cluster center count of 2048, or exhausting the graph work cap each produce an unresolved outcome rather than a negative certificate.

The recorded execution.

The complete run consists of 100 program invocations:

- one pass over the seed cover;
- four shards of the completion stage;
- two refinement passes, a mid one and a fine one;
- 88 batches of the initial exclusion pass;
- one fallback refinement pass;
- four final exclusion batches.

The initial exclusion pass leaves a nonzero count on some batches: the residual candidates that the bin and graph tests could not reject. They are handed to the fallback, whose own batches then feed the four final exclusion batches. Recorded counts and hashes accompany the whole run.

For every initial exclusion process the recorded status is 0 exactly when its unresolved count is zero, and 1 exactly when that count is positive. The four failure categories — uncapped graph candidate, graph-step cap, bin overflow, and clustering failure — sum to that count, and none of them is an exclusion. Every nonempty batch reaches the fallback, sorted by locale, including batches whose locations differ from one run to the next. The final exclusion pass uses all four modulo-4 parts of the fallback output. Had the initial pass left no unresolved record, the fallback and the final pass would have been unnecessary; that shortcut was not taken.

Fifteen initial exclusion batches left unresolved records, and together they carry all 163 unresolved records. There were no initial bin overflows, clustering failures, or uncapped graph candidates. There were no final failures of any category. All 100 program invocations are therefore accounted for: one cover pass, four completion shards, two refinement passes, 88 initial exclusion batches, one fallback pass, and four final exclusion batches. The fifteen nonzero exits are fully resolved by that fallback chain. No rejection threshold, graph cap, refinement width, or mathematical input was altered, and no additional refinement level was inserted.

For scale, the measured wall times, including polling and validation, were 621.59 seconds for the cover, 2020.32 seconds for the four parallel completion shards, 1071.49 and 407.45 seconds for

the two refinement passes, and 300.23, 8.12 and 36.27 seconds for the initial exclusion, the fallback and the final exclusion. The outer supervisor measured 4470.33 seconds for the complete pipeline. These are observations on this platform, not native CPU times or running-time guarantees. No stage hit its wall deadline, and no retry or resource limit was raised.

The execution archive preserves the numerical data and records of this complete run, with internal path and host fields redacted: sources and binaries, center and record files, all 88 initial exclusion batches and their unresolved records, fallback data, every final unresolved record, stdout and stderr, commands, exits, platform information, and validation records. Inside it one record describes the whole run, one record per stage describes that stage, and one record per child describes that child. Rebuild and execution instructions accompany the archive. Archived native binaries document this run; they are not a portability claim.

The certificate depends only on this newly completed chain and the counts recorded alongside it. No earlier execution, diagnostic sample, or intermediate timed measurement substitutes for any stage of this full run.

The finite-execution certificate.

Theorem 54 (Finite-execution certificate)**.** *Fix the four recorded programs and the arithmetic conditions of the certified arithmetic. Suppose the recorded invocations meet every terminal condition listed above, together with the record provenance and batch checks stated there. Then four mutually unbiased orthonormal bases in* $\mathbb{C}^6$ *do not exist.*

Proof. Suppose four such bases exist. Normalize the first to the standard basis and write the second as $H/\sqrt{6}$.

Seeds. Proposition 13 places the seed coordinates of an equivalent exact Hadamard matrix in one of the generated seed balls. All A shards together process that seed.

Stages A and B. Proposition 39 therefore places an equivalent exact solution in a coarse Record. Each equivalence preserves extendibility by two further bases (Lemma 5). On that Record the refinement stage replaces the ball by a finite atlas of children together with any unresolved parent; Proposition 44 gives containment. Since the run has no remaining bad parent at the end of the initial C stage and no bad record at the end of the final C stage, the record is either covered by a completed child at its own precomputed local certificate, or refined into a further record that is similarly covered.

Stage C. Proposition 53 then rules out two extra bases in every such local ball. The exclusion run leaves no unresolved record at the end, by terminal condition 5 together with the terminal conditions of the fallback chain. Therefore every individual pointer to the Hadamard solution lies in a phase ball that the exclusion stage rules out.

Contradiction: the assumed family would induce a Hadamard displacement that lands inside some ball of the final cover, but each such ball has a completed negative certificate. Hence four such bases do not exist. □

The certificate therefore has three independent components: the finite-arithmetic premises of the certified arithmetic, the finite invocations and terminal conditions listed above, and the mathematical chain of the preceding stages. A change to any one of them — different build flags, different terminal conditions, or a different proof of one of the local lemmas — would require a fresh execution of the affected stage, but the logical structure of the certificate is unchanged.

The theorem and its proof complete the computer-assisted upper bound.

Conclusion of the proof.

The proof of Theorem 1 is now complete. The overall argument is the following chain of strictly finer covers, each preserving every exact candidate solution.

1. By Lemma 5, every complex Hadamard matrix, up to allowed symmetries, has its nine-phase seed inside one of finitely many closed balls (Proposition 13).

2. For each such seed ball, the possible completions form a $U(2)$-orbit; the $U(2)$ cells are divided into finitely many pieces such that each piece either is excluded by a certified necessary condition or is covered by a small phase ball (Proposition 39).

3. Each Phase ball is subdivided into smaller children, and the original balls are covered by new children together with any unresolved bad parents (Proposition 44).

4. Each small phase ball is ruled out by the C-stage graph test on independent vector balls and pairwise tests (Proposition 53).

The recorded execution of the pipeline meets every terminal condition listed above, so the finite-execution certificate Theorem 54 applies and yields the negative upper bound. The lower bound is Lemma 2; the two together give $N(6) = 3$.

The argument can be summarized in four sentences: cover the space of complex Hadamard matrices; replace the cover with smaller and smaller open covers that still contain every exact solution; within each small cover, exclude two further mutually unbiased bases by the graph test; and check the whole chain by an actual computer run with certified arithmetic.

The historical and mathematical antecedents surveyed above are complementary; the combination needed for the theorem above is new. In particular, the present proof covers every second-basis Hadamard matrix directly, rather than running a certified search inside a presumed family.

Certified arithmetic for the recorded execution.

This appendix records the outward error estimates that back each inequality quoted in and . Every certified rejection is a strict violation of an outward bound computed under the primitive-operation model below. Nothing here assumes that a floating-point center is an exact solution, and no step relies on the observed size of an error over a specific input.

Primitive-operation model and unit roundoff.

The primitive-operation model is the binary64 specification of IEEE 754 [12, Sections 3.4, 3.6, 4.3, 5.4.1 and 7.5]. The accumulation estimates use the usual product-of-rounding-factors argument [11, Section 2.12 and Lemma 3.1], with the explicit additive underflow allowance retained here.

The arithmetic premises are separate, correctly rounded binary64 addition, subtraction, multiplication, division, and square root in round-to-nearest mode, with binary64 evaluation of intermediate results. The build uses `-ffp-contract=off` and does not use fast-math or any reassociation that changes these operations. Subnormals are not flushed to zero, and denormal inputs are not treated as zero. Floating exceptions are masked, so an unsuccessful heuristic calculation can produce an infinity or NaN and be rejected by its subsequent checks. The relevant integer types, numeric locale, and file representation are specified in .

The function `environment` sets and checks nearest rounding and checks the binary64 properties of `double`. The A/B/C entry points also inspect MXCSR, rejecting a non-nearest mode, DAZ, FTZ, or unmasked exceptions. The same no-DAZ/no-FTZ environment is required for `covout`, although its entry point does not itself inspect those bits. The default-nearest premise also applies to constants and to the global basis constructed before `main`. These are explicit arithmetic premises, not a claim that a source-level static assertion formally verifies every compiler transformation. The recorded build checks and their limits are described in .

The functions `nearbyint` and `floor` choose integers at finite bounded arguments. The particular integer chosen at a boundary is permissible wherever the proof subsequently uses that integer lift. No accuracy premise is imposed on `atan2`: Candidate checks the unit exponential of the phase

it proposes. Decimal literals are either used as exact stored parameters, or allowed a relative translation error of at most $5 \cdot 10^{-16}$ when compared with a displayed decimal.

Put

$$u = 2 \cdot 10^{-16}, \qquad \tau = 10^{-300}, \qquad \gamma_m = 2.1 \cdot 10^{-16}(m+1). \tag{129}$$

For a correctly rounded basic operation with finite exact result z at the scales below,

$$|\,\mathrm{fl}(z) - z| \le u|z| + \tau. \tag{130}$$

The true binary64 relative unit roundoff is smaller than u, and the additive allowance is much larger than half the least positive subnormal. Overflows are excluded for the certificate evaluations by the scale bounds proved below; they need not be excluded in an untrusted solve.

Lemma 55 (Accumulation and norm bounds)**.** *For $m \le 2000$, a dot product on exact stored inputs, formed as separate products and successive additions, satisfies*

$$\left| \mathrm{fl}\left(\sum_{i=1}^{m} a_i b_i \right) - \sum_{i=1}^{m} a_i b_i \right| \le \gamma_m \sum_{i=1}^{m} |a_i b_i| + 4(m+1)\tau. \tag{131}$$

An initial addend is included in the absolute sum and increases the small operation-count constant by one. A product of three factors instead of two increases that constant by one more. For the square-sum/square-root norms used here,

$$|\mathrm{fl}(\|a\|) - \|a\|| \le 2.2 \cdot 10^{-16}(m+2)\|a\| + 10^{-140}. \tag{132}$$

If the computed components approximate other components, their Euclidean difference is added to this bound.

Proof. Iterating $|z_k| \le (1+u)|z_{k-1}| + u + b\tau$ for a length-m dot product with the standard addition/rounding analysis (Higham [11, Lemma 3.1]) gives an interpolation of the form $(1+u)^m \sum_{i=1}^m |a_i b_i|$ plus a $O(m\tau)$ term. The bound $\gamma_m = 2.1 \cdot 10^{-16}(m+1)$ is a conservative choice: for $m \le 2000$ one has $(1+u)^m - 1 \le mu \le 2m \cdot 10^{-16}$, which is smaller than γ_m. Direct substitution of the accumulated additive terms into the recurrence produces the τ bound. The norm inequality is the same argument applied to the square-sum and then to the correctly rounded square root, with $m+2$ factors and a final 10^{-140} additive allowance. □

Pi and exponential evaluations.

The constant π is stored as the correctly rounded binary64 literal. Its absolute error is at most 10^{-16} and its relative error is at most u. The constants $\sqrt{6}$ and its reciprocal are formed by the indicated square-root and division; $2 \cdot 10^{-15}$ is an ample absolute error bound for each.

The source evaluates cosine through degree 18 and sine through degree 19. In the jth nonconstant term, the error in $-y^2$ is reused j times, and there are j further multiplications and j divisions. The small integer denominators are exact. Its relative error is therefore at most $(1+u)^{3j} - 1 < 5.5 \cdot 10^{-15}$ for $j \le 9$. For additive underflow errors, compare with a recurrence using the same multiplicative rounding factors but zero additive errors. Its terms have magnitude below two, and each recurrence contracts the previous additive error by a factor less than $1/2$: the numerator factor has magnitude below $.786^2$ and the smallest denominator is two. Errors from that factor, the product, and the division give the conservative recurrence $E_j \le E_{j-1}/2 + 4\tau$, $E_0 = 0$. Thus $E_j < 8\tau$; after both coordinates' nine additions their total contribution is below 10^{-295}, including when y^2 or a later term underflows. The absolute even-term sum at .786 is below 1.4, its nonconstant part below .4;

the odd-term sum is below 1, its part after y below .2. For the nonconstant even terms the successive ratio after the first is at most $.786^2/12$, and for the odd terms after y it is at most $.786^2/20$. Their geometric majorants give the claimed absolute sums without evaluating a transcendental function. The first omitted cosine and sine terms are below $3.33 \cdot 10^{-21}$ and $1.25 \cdot 10^{-22}$, respectively; both are below 10^{-16} even in the worst scale considered here. Thus

$$|\operatorname{fl}(e^{iy}) - e^{iy}| \le 10^{-14} \tag{133}$$

for all real y with $|y| \le 20$ and no overflow or underflow.

For $z = a + bi$ with $|a|, |b| \le 20$, the pair $(\operatorname{fl}(\cos b), \operatorname{fl}(\sin b))$ is multiplied by e^a; the pair error and the multiplication error are each below $2u$ in relative terms, and the total error is below 10^{-14} in absolute value at the scales used here (bound:). All relied-on arguments satisfy the range condition. Seed and dyadic unitary phases are bounded near $[0, 1]$; `gen` checks $|p_j| \le 4$; Candidate checks $|P_j| \le .51$ before its exponential; C validates $|p_j| \le 4$ and checks ball phases of magnitude at most two. The largest C difference has magnitude at most six.

Dephasing and projection.

For $|p_j| \le 4$, the three operations in unwrapped dephasing $p_{ij} - p_{i0} - p_{0j} + p_{00}$ have entry error below $8 \cdot 10^{-15}$; their intermediate magnitudes are at most 8, 12, and 16. Subtracting the integer selected by `nearbyint` gives error below 10^{-14} against the exact dephase minus that same integer. A wrapped difference of two such arrays has error below $3 \cdot 10^{-14}$ per entry against the ideal difference with the combined selected integers. The one-sided Candidate version, whose other array is already a stored dephased phase, has the same bound. These comparisons do not assume identical rounding choices for an independently formed ideal phase.

For a stored wrapped array, $|z_j| \le 1/2$. Its row and column sums have error below $4 \cdot 10^{-15}$, and its length-36 grand sum below $1.4 \cdot 10^{-13}$. Dividing by 6 or 36 and combining the means gives projection-operation error below 10^{-14} per component. Projection contracts the prior input error. Consequently the computed projected difference has Frobenius error below 10^{-11}. All distance square and norm errors at the successful cutoffs below are negligible against 10^{-7}.

Thus `Atlas::find`'s returned $\sqrt{s} + 10^{-7}$ is an upper distance for the selected ideal projected lift. A merge uses radius $r + \text{distance} + 10^{-8}$ and checks a cap with $2 \cdot 10^{-8}$ slack. Its sum-rounding error is below 10^{-14}. The same arithmetic applies to `assign` and to the local sphere-radius additions. Searching only some buckets or transforms may miss a merge; it cannot certify a merge that fails these tests.

Seed filtering and seed-ball reuse.

Here $n = 112$ and the stored w approximates $1/n$. Midpoint-gap calculations contain only small integer arithmetic and a few products/subtractions; (130) bounds their total error, including the comparison threshold, by 10^{-14}. Their slack is 10^{-8}. The orientation filter is integral.

For a list of five varying phases, the computed residual c in `sumok` has component error below $2 \cdot 10^{-13}$. Indeed its five exponential errors, after division by 2π, total less than $4.8 \cdot 10^{-14}$ in complex modulus, and the divisions, additions, and constant error cost less than $3 \cdot 10^{-15}$. The sharper componentwise bound $1.1 \cdot 10^{-13}$ will also be useful. Each test direction is regarded as an exact stored pair; its norm is less than 1.1. The five column perturbations, the residual perturbation, and ordinary arithmetic in `chk` cost together less than 10^{-11}. Its remainder coefficient approximates $5\pi/(4n^2)$ within 10^{-15}. The actual 10^{-8} comparison addition therefore preserves every feasible list.

In `supdual`, all proposed entries are checked by $|v_a| \le 10^4$, with a negated comparison that also rejects NaN. They are then exact stored test coefficients. The four right-hand-side errors against

$-nc$ are below $3 \cdot 10^{-11}$ each. The signed term consequently has input error at most

$$2 \sum_{a=1}^{4} |v_a| \, 3 \cdot 10^{-11} \le 2.4 \cdot 10^{-6} < 2.5 \cdot 10^{-6}.$$

Its absolute term sum is below 10^7, so its arithmetic error is below 10^{-7}. Each of the nine row-dot residuals has error below $4 \cdot 10^{-9}$: its four input contributions are at most $4 \cdot 10^4 \varepsilon_{\exp}$, and its short dot costs less than 10^{-10}. Each pair norm has error below 10^{-10} by (132). The remainder coefficient uses 1.251 instead of 1.25, which leaves positive slack after its own rounding. The final absolute scale is below $2 \cdot 10^7$. Summing these contributions shows that the ideal support expression is at most the computed proposal plus 10^{-4}. The actual addition is 0.01. If no usable proposal improves the initial 10^{10} bound, that initial bound already exceeds the trivial support.

Integer differences lie in $[-111, 111]$, including those stored in `int8_t`; all their squared/absolute sums are exact and small. The comparison radius is

$$\texttt{RN2} = \mathrm{fl}((\mathrm{fl}(.032 \cdot 112))^2) - .0001$$

with the source's indicated operation ordering. The total possible error before the subtraction is below 10^{-12}, so this is smaller than the ideal $(.032 \cdot 112)^2$. This reasoning requires the `pow(x,2)` calls in `covout` to be lowered to binary64 multiplication, as checked for the fresh build documented in . It is not a claim about an arbitrary library implementation of `pow`. The other square-power value T, bucket reach, and integer cuts only select reuse attempts; missing an attempt does not discard the current bin pair. A newly inserted midpoint needs radius only $1.5/112 < .032$. Finally, the nine division-rounded seed phases differ from their rational midpoints in norm by less than 10^{-14}, which is absorbed when the A seed radius is increased to .0320001.

Frames, comparison blocks, and A exclusion tests.

All ideal quantities in this subsection use exact unit exponentials at the specified centers, and the same coordinate indices selected by the code. We never need the selected coordinate to maximize the *ideal* projection score.

For `frame`, the four-term sum S has error below $2.5 \cdot 10^{-13}$; its squared modulus has error below $2.2 \cdot 10^{-12}$, and its modulus below $2.6 \cdot 10^{-13}$ by (132). If the stored $m = \sqrt{4 - |S|}$ passes $m > 1.25$, the ideal $m > 1.24$. Using $|\sqrt{a} - \sqrt{b}| = |a - b|/(\sqrt{a} + \sqrt{b})$ at these lower bounds gives errors below $3 \cdot 10^{-13}$ in m and top $= \sqrt{4 + |S|}$. In particular $16 - |S|^2 > 6.1$; its error is below $2.3 \cdot 10^{-12}$, so its reciprocal has error below $7 \cdot 10^{-14}$. The two pseudoinverse numerators have modulus at most eight and error below $5.1 \cdot 10^{-13}$. Division/scaling then gives

$$|\Delta G_{ij}| < 10^{-12}, \qquad |G_{ij}^{\mathrm{ideal}}| < .81, \tag{134}$$

the second inequality following from the lower singular bound.

The first constant basis column is exact. The next column has numerator error below $1.3 \cdot 10^{-13}$ and denominator $\sqrt{4 - |S|^2/4} > 1.5$ with error below $2 \cdot 10^{-13}$. Its ideal entries have modulus at most one, so the quotient error is below $4 \cdot 10^{-13}$. For each of the remaining two columns, use the actual selected coordinate projection q. In the first such projection the earlier two products give entry error below $9 \cdot 10^{-13}$, and its norm error is below $2 \cdot 10^{-12}$. The accepted computed norm is greater than .449999, so the ideal projection is nonzero and normalization gives entry error below $7 \cdot 10^{-12}$. In the last projection the extra column-product error gives entry error below $1.6 \cdot 10^{-11}$ and norm error below $3.5 \cdot 10^{-11}$. The same denominator check gives entry error below $1.2 \cdot 10^{-10}$. This argument inductively constructs the ideal orthonormal complements with the selected coordinates.

The two-product kernel projector has entry error below $3 \cdot 10^{-10}$; multiplying two complement entries and $\sqrt{6}$ gives a W entry error below $8 \cdot 10^{-10}$.

The following table records the rest of the propagation, with the source array names. Its errors include ordinary operation rounding.

Quantity	Absolute error per entry	Ideal scale used
`base`	10^{-11}	two factors of modulus at most 2 times G
`fixed`	$2 \cdot 10^{-9}$	projector times a factor at most 1.62
`bm`, `cm`	$2 \cdot 10^{-12}$	unit entry times G
`uc.U`, angular derivatives	$2 \cdot 10^{-13}$	products of unit phases/trig entries
`K`, `D`	$4 \cdot 10^{-9}$	four W products and base
`mag`	$4 \cdot 10^{-8}$	$\lvert D\rvert < 4.6$
$\sqrt{\texttt{mag}}$	$5 \cdot 10^{-9}$	norm Lipschitz
`jac`	10^{-8}	ideal modulus at most 6
flat real J	$1 \cdot 10^{-7}$	dot of D and jac
flat c	$5 \cdot 10^{-8}$	$(\lvert D\rvert^2 - 1)/2$
pre-equation c	$4 \cdot 10^{-13}$	short unit-exponential sums
pre-equation coefficients	$6 \cdot 10^{-14}$	unit exponentials

The horizontal-projector identity.. An entry of the exact double-centering projector is the product of two centering-matrix entries. The computed `PI` differs by less than 10^{-15} per entry. In the matrix L_P, each row has only 12 nonzero entries, with error below $1.3 \cdot 10^{-13}$ from the edge products. Forming `LP` gives error below $5 \cdot 10^{-12}$ per entry against $L_P\Pi$; the ideal entry magnitude is at most 8.4. In the residual $\Pi - TT^* - ML_P\Pi$ each computed entry then has error below 10^{-9}. The propagated LP error is at most $30 \cdot 2 \cdot 5 \cdot 10^{-12} = 3 \cdot 10^{-10}$; the remaining PI, T-product, and short accumulation errors fit in the remainder of the budget, with absolute sum below 560. The 1296 entry errors have Frobenius total below $3.6 \cdot 10^{-8}$. Norm accumulation is below 10^{-8} even with residual norm 20161. Thus the actual $2 \cdot 10^{-6}$ addition in `id` suffices.

Root certificates and normal constants.

The tangent operator has the certified bound

$$\|T\|_{\mathrm{op}} \le \sqrt{1 + \texttt{ort}} < 1.00001, \tag{135}$$

since T is a genuine Root. The evaluation of $T^*T - I$ for the stored T (each $|T_{ja}| \le 1.01$) yields $\texttt{ort} < 10^{-5}$; the evaluation of $(I - \Pi)T$ yields $\texttt{hor} < 10^{-5}$; the evaluation of $F(P)$ yields $\texttt{froot} < 10^{-6}$.

For the auxiliary maps M: the source forms $M = H_V L_V^*$ with at most 25 columns of H_V and L_V, each checked to magnitude at most 2. Absolute row sums of the column Grams of H_V and L_V give upper bounds on $\|H_V\|$ and $\|L_V\|$; their product overestimates $\|M\|$. Adding the $2.1 \cdot 10^{-11}$ Frobenius bound on the difference $H_V L_V^* - M$ (where M denotes the accumulated stored product) gives a valid `mnorm`.

The two candidate choices for C are $\sqrt{8}\|M\|$ and $\sqrt{6}\|\mathcal{S}\|$; the source takes the smaller of upper bounds for these two quantities. Each is a valid πb in Lemma 27 by Lemma 28. The root evaluation thresholds are $\texttt{val} < 10^{-18}$ for the overall pass, $\texttt{ort} < 10^{-5}$, $\texttt{hor} < 10^{-5}$, $\texttt{id} < 10^{-4}$ after the $2 \cdot 10^{-6}$ addition, and $\texttt{mnorm} \cdot \texttt{froot} \le \texttt{n0}$.

Candidate conversion and ellipsoidal coverage.

These coverage estimates are needed only after a successful Candidate preparation, at final unitary-grid levels $N_u \ge 64$.

Let $m_j = |D_j|$ denote an ideal comparison-block magnitude. The stored magnitude test $\sqrt{\texttt{mag}_j} > .15$ implies $m_j > .14999998$. Normalizing D by its computed magnitude has unit-direction error below

$$\frac{2 \cdot 4 \cdot 10^{-9}}{.14999998} + \text{norm/division rounding} < 5.5 \cdot 10^{-8}.$$

Candidate additionally checks a computed chord smaller than 10^{-7} between this normalized value and the exponential of its proposed phase. With (133), the true unit chord error is less than $1.56 \cdot 10^{-7}$. If a circular phase distance in cycles is $v \in [0, 1/2]$, its chord is $2\sin(\pi v) \geq 4v$. Across 16 entries the phase-center correction therefore has norm below $1.56 \cdot 10^{-7}$, which is below the $2 \cdot 10^{-6}$ allowance used in . The nine seed entries use exact stored seed centers and an exact integer subtraction.

For the D-position coefficients X, the normalized-direction error times the Jacobian bound six, plus the 10^{-8} Jacobian error and short-dot rounding, is below 10^{-6} per entry. The other used X entries are exact. The `mi` subtraction of 10^{-6} therefore gives a positive magnitude lower bound greater than .148, and $\texttt{invmag}_j = 1/(\widehat{m}_j - 10^{-6}) + 10^{-9}$ is an upper bound for $1/m_j$, below 6.8.

The flat-sum error is below 10^{-6}, so the 10^{-5} addition in `slack`, followed by its additional 10^{-10}, gives the required nonnegative weighted chord-square upper bound. In the passed branch $q < 3.8$ and $1 - q/4 > .049$. The arithmetic in dividing `slack` by `mi` has error below 10^{-12}: `slack`< 4.1 and the denominator is above .148. The ratio $(1 - q/4)^{-1/2}$ has error below 10^{-12} before its 10^{-7} addition; the denominator lower bound bounds the derivative, and $q/4$ is a binary scaling. The remaining positive products/roots and the division by $(2\pi)^2$ in `d2` each have error below 10^{-12} before their 10^{-8} additions. In particular the stored quantities are upper bounds with

$$\texttt{arc} < 8.8, \qquad \texttt{weighted} < 10, \qquad \texttt{invmag}_j < 6.8. \tag{136}$$

Tangent offsets and external errors.. The computed `off` approximates $2\pi T^* \Delta$ within 10^{-9} in norm, by (135), the projected-difference error, and short-dot rounding. The error in the whole X array is at most $\sqrt{208} \cdot 10^{-6}$ in Frobenius norm; there are 13 columns and 16 affected positions. Multiplication by T^* and the less than $2 \cdot 10^{-13}$ rounding of each dot give

$$\|\texttt{AB} - T^* X_{\text{ideal}}\| < 1.5 \cdot 10^{-5}. \tag{137}$$

The parameter-vector norm is below $\sqrt{.203^2 + 4(\pi/64)^2} < .226$, so substitution of stored AB costs less than $3.4 \cdot 10^{-6}$ radians. Formation of the corner offsets adds less than 10^{-11}, using coefficients of magnitude below 25. The same bound applies to convex combinations of these corners.

For the second covariance, use stored `weighted` and `invmag`, with rational $\beta = .8$. Each D-weighted dot has absolute sum less than $16(1.00001)^2 \cdot 6.8 < 110$. Its two-product-per-term accumulation has error below $5 \cdot 10^{-13}$ by (131) with one extra factor. The multiplying coefficient is below 225. Coefficient translation and the subsequent arithmetic, even at absolute intermediate scale 26000, give total entry error below 10^{-9}. Both G arrays are explicitly stored symmetrically. The added $10^{-7} I$ dominates k times either entry-error bound for $k \leq 10$. Hence the exact stored G is positive semidefinite and dominates the required covariance.

The stored spectral upper bound.

Lemma 56. *For an exact stored symmetric $k \times k$ matrix G with $k \leq 10$ and $|G_{ij}| \leq 10$, `spectralUpper` returns an upper bound for $\|G\|$.*

Proof. An absolute row sum has error below $3 \cdot 10^{-13}$. The code adds 10^{-10} to their maximum. For its other bound, put $m = \max_{ij} |G_{ij}|$ and $A = \|G^2\|_F$. Because one diagonal sum of squares

contains a term m^2, $A \ge m^2$. Then $|\,\mathrm{fl}(\sqrt{\texttt{fro4}}) - A| < 4 \cdot 10^{-13} A + 10^{-139}$ by dot-product and norm rounding. Taking $\sqrt{A}$ costs less than $4 \cdot 10^{-13}\sqrt{A} + 10^{-69}$, and since $\sqrt{A} \le 100$ the total possible downward error is below 10^{-10}. The code adds $2 \cdot 10^{-7}$. The exact $\sqrt{\|G^2\|_F}$ bounds $\|G\|$. Both returned candidates are upper bounds, so their minimum is also an upper bound, and its value is nonnegative and at most 100.1. □

The `ell` guard checks symmetry and $|G_{ij}| \le 10$, and each offset component has magnitude at most ten. Its invalid return value 100 cannot certify a cell: after conversion to cycles it exceeds 10, so the associated bootstrap returns its nonsuccess sentinel. A different valid bound may still succeed. For valid data the initial triangle bound is $\sqrt{\texttt{mx}} + \sqrt{\texttt{gu}} + 10^{-8}$, with `mx` the maximum squared offset norm. Its norm error is below 10^{-13}, and the triangle bound is smaller than 43.

For each proposed resolvent parameter γ: $0 < \gamma < 418$, and

$$\texttt{gap} = \mathrm{fl}(\gamma - \texttt{gu} - 10^{-9}) \le \gamma - \lambda_{\max}(G).$$

Only gaps at least 10^{-6} are used. Taking max with zero and over all offsets gives `high`. The ellipsoid formula uses $\sqrt{\gamma(1 + \texttt{high})} + 10^{-6}$; the added 10^{-6} and the whole-candidate error bound 10^{-12} dominate at the used scales.

The base grid and the outside test.

Use the stored positive width w_B as an exact grid parameter. The allowed range is $10^{-4} \le w_B \le 0.05$. The stored grid radius $E = \sqrt{1 + \texttt{ort}_P}\, R + 10^{-7}$ exceeds the true tangent radius by more than $0.999 \cdot 10^{-7}$. For $L = \lceil E/w_B + 0.500001 \rceil$, the quotient scale is at most 1261 and its error is below 10^{-12}. The 0.000001 excess over the half-cell term is much larger, so the integer base range covers $[-E, E]$ in every coordinate. In particular $L \le 1262$.

Exact base centers have magnitude below 0.203 per coordinate. Their computed products have error below 10^{-16}. Halving a stored width is exact here; after at most six subdivision levels, each computed center differs from the intended exact dyadic child center by less than 10^{-15} per coordinate. The tangent-restriction identity $\|T^*\delta\| \le \|(\Pi T)^*\delta\| + \|(I - \Pi)T\|\,\|\delta\|$ is used to reduce the parent-distance test to a single comparison against E_P, without loss elsewhere.

Single-vector filters and vector-ball construction.

The C stage validates its Root and $0 \le r \le 0.03$. At every bin resolution $M \ge 6$, the halfwidth, Euclidean bin radius, one-norm bound, and infinity-norm bound are formed as

$$a = 0.5/M + 10^{-12}, \quad X = \sqrt{5 - 1/6}\, a + 10^{-12},$$
$$x_1 = 5a + 10^{-12}, \qquad x_\infty = 1.5a + 10^{-12}.$$

Their operations have error below 10^{-14}, and the 10^{-12} additions are upward. They satisfy $X < 0.19$, $x_1 < 0.42$, and $x_\infty < 0.126$. The stored tangent and normal bounds

$$r_t = \sqrt{1 + \texttt{ort}}\, r + 10^{-9}, \qquad r_n = \texttt{n0} + \texttt{id}\, r + C r^2 + 10^{-9}$$

are upper bounds, with $r_t < 0.030001$ and $r_n < 0.019$. The square-root/product and positive-polynomial errors are below 10^{-13}. The constant $K = 1.00001$ exceeds the operator bound regardless of literal translation; the Root residual gives

$$\|e(P)/\sqrt{6}\| \le \sqrt{1 + (2\pi\sqrt{2}/6)\, 10^{-6}} < 1.00000075 < K.$$

Storage and elementary arithmetic are bounded directly: the phase magnitude is below 4.0; the six-term S error is below $6 \cdot 10^{-14}$; $|\,|S|/\sqrt{6} - 4\,| < 10^{-13}$. The arrays J and B have errors

below $3 \cdot 10^{-11}$ and $4 \cdot 10^{-11}$. The rounded constants $a, X, I_M, L_M, 1/\sqrt{6}, (2\pi)^2$ each carry unit relative error u, and the two required products and the sum carry at most $3u$ relative. Each bin-face primitive check has constant $2 \cdot 10^{-7}$ above $r_n\|J\| + (r_t/4)$, using $|x_j| \le a$; its threshold is 10^{-6}. The combined bound uses the same $2 \cdot 10^{-7}$. The nested squared-magnitude crossing band is $2 \cdot 10^{-7}$; the squared increment is $7 \cdot 10^{-7}$.

The directional single-vector support.. The heuristic solve is accepted only with finite coordinates and a maximum magnitude between 10^{-100} and 10^{100}. After division by that maximum, regard the resulting stored $|y_i| \le 1$ as the test coefficients. Define exact stored arrays

$$\begin{aligned} a_y &= \sum_i y_i c_i, & h_a &= \sum_i y_i B_{ia}, \\ z_j &= \sum_i y_i J_{ij}, & \widetilde{z}_j &= \sum_i y_i D^{\mathrm{re}}_{ij}, \\ N_J &= \sum_{i,j} y_i^2 J_{ij}^2, & N_S &= \sum_i y_i^2 |S_i|^2, \\ M_S &= \max_i y_i^2 |S_i|^2, & m_y &= \max_i |y_i|, \\ M_z &= \max_j (z_j^2 + \widetilde{z}_j^2). \end{aligned}$$

The support compared with $|a_y|$ is the stored evaluation of

$$\begin{aligned} & a \sum_{j>0} |z_j| + r_t\|h\| + r_n\sqrt{N_J} \\ & \quad + \pi\Big[m_y(KX + r)^2 + \min\{\sqrt{M_z}\,X,\ K\sqrt{N_S}\,x_\infty\}X + 2\frac{\sqrt{N_S}}{\sqrt{6}}Xr + \frac{\sqrt{M_S}}{\sqrt{6}}r^2\Big]. \end{aligned} \tag{138}$$

This is the necessary support bound of the Exclusion stage. Its coefficient errors against ideal arrays satisfy

$$|\Delta a_y| < 7 \cdot 10^{-12}, \qquad |\Delta z_j|, |\Delta \widetilde{z}_j| < 7 \cdot 10^{-12}, \qquad |\Delta h_a| < 4.3 \cdot 10^{-11}.$$

The first two use six input errors and a short dot; the last uses six B errors and the same dot estimate. The two bracketed products cost less than 10^{-14} each; the short absolute sums and the normalization by $m_y \ge 10^{-100}$ cost less than 10^{-15}. Squares and square-root evaluation at the used scales cost less than 10^{-15}. Three times either dot error is below $3 \cdot 10^{-11}$. Thus the certified support contribution is within 10^{-11} of the ideal expression inside π, and the whole error against the ideal support is below 10^{-10}. The code adds $2 \cdot 10^{-7}$.

The joint coefficient support.

The joint test is separated over the three tangent pieces and has at most $6+6+2 = 14$ rows. The proposed direction is checked finite and normalized as before, so its stored entries have magnitude at most one.

The constant pairing has error below $1.5 \cdot 10^{-11}$. Each accumulated shared-Hadamard coefficient has error below $9 \cdot 10^{-11}$: it receives at most twelve B input errors of $7 \cdot 10^{-12}$ plus short-dot rounding. Each vector-phase coefficient, including its edge contribution, has error below $9 \cdot 10^{-12}$: six single-vector entries and at most two edge entries each contribute at most $12u$. Combining, the ideal support is at most the computed support plus $3 \cdot 10^{-10}$. The code adds $3 \cdot 10^{-7}$.

The final comparison allowance of $3 \cdot 10^{-7}$ is monotone: NaN and infinity are rejected by the direction checks, so only finite proposals enter. Enlarging the proposed direction or its stored coefficients can only raise the computed support, so the comparison remains a rigorous upper bound at the recorded threshold.

Finite paths and NaN handling.

The preceding estimates apply to bounded finite certificate evaluations. Exceptional heuristic values cannot bypass those preconditions.

1. In the seed cover, all direct lists come from finite integer midpoints and `expair`. The dual solver's entries must pass the negated bounded comparison $|v_a| \le 10^4$; NaN and infinity fail it. A failed proposal leaves the large safe support bound. Gap/hash/reach cuts select only possible reuse attempts.

2. For A, a failed frame normalization makes setup fail and the entry point records an unresolved seed ball. It does not certify the seed ball empty. The passed frames, fixed radius, and dyadic unitaries give finite bounded block coefficients. The potentially ill-conditioned solves only propose directions; the final `trial::certificate` checks every direction entry and its maximum before use.

3. The snapping iteration calls `gen`, which rejects nonfinite or out-of-range phases. Root certification requires `val` $< 10^{-18}$, checks proposed eigenvalues finite, checks HV/LV before using them, and checks T before its certificate products. Those bounded arrays make the products finite. The optional tensor change of basis is range-checked; an unusable one cannot lower the previously valid C. The final conjunction of strict Root thresholds also rejects NaN.

4. In B, failed cells are retained as bad parents; they do not enlarge the covered region. The marker file is replaced only after a successful atlas update, and the final count is the terminal cover size. A failed temporary write records an unresolved parent.

5. The cluster output is checked as in at $|v_j| \le 2$ and $0 \le s < .1$. The two edges share the distance bound $r_e = s_u + s_v$. If the direct tests reject an edge, its weight is zero with infinity cost. Otherwise the stored D_E and q_E are finite. The joint coefficient guard from then rejects any failed proposal and keeps the corresponding previous bound. A failed proposal cannot certify a pair excluded.

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